UNDERGRADUATE THESIS REPORT

Optimal Aircraft Maintenance Task Packaging for a Single A320 Aircraft: An Integrated Mixed-Integer Linear Programming (MILP) Application

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Abstract

Ensuring aircraft airworthiness and optimizing maintenance planning is a critical challenge in aviation engineering. This report details the development and application of a **Mixed-Integer Linear Programming (MILP)** model designed to optimize the packaging of maintenance tasks for a single aircraft. The research employs an outcome-oriented design philosophy, where the necessity for a traceable, verifiable, and discrete output drives the choice of the MILP methodology.

The study outlines a methodology that includes a detailed review of engineering contexts (airworthiness, RCM), formal problem characterization as an NP-hard combinatorial problem, and the design of a custom software pipeline for implementation. A single-aircraft case study, utilizing data from a specific MRO facility, was used to validate the model.

Results demonstrate that the MILP model successfully minimized total maintenance costs while achieving full compliance with all operational and regulatory constraints. When benchmarked against a baseline heuristic model, the MILP approach proved superior, yielding a **[5%]** saving in operational costs and **20%** saving in man-hours [**depending on which result use and shown in Chapter 4**] for the case study. The findings confirm that rigorous optimization is feasible and valuable for tactical maintenance planning. The report concludes by proposing a roadmap for future work, focusing on integrating this validated MILP model into hybrid metaheuristic framework to address large-scale, fleet-wide scheduling complexity.

Keywords: Aircraft Maintenance Planning, Optimization, Task Packaging Problem, Mixed-Integer Linear Programming (MILP), Airworthiness, Reliability Engineering.

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I Kadek Surya Geryandika

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Abbreviation

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Terms	Acronym
AD	Airworthiness Directives
ALS	Airworthiness Limitations Section
API	Application Programming Interface
CAMO	Continuous Airworthiness Maintenance Organization
DEM	Dynamic Execution Model
DMC	Direct Maintenance Cost
EASA	European Union Aviation Safety Agency
FAA	Federal Aviation Administration
FMEA	Failure Mode and Effect Analysis
GUI	Graphical User Interface
IMC	Indirect Maintenance Cost
LP	Linear Programming
MH	Man-hours
MILP	Mixed-Integer Linear Programming
MPD	Maintenance Planning Document
MRO	Maintenance, Repair, and Overhaul
MRB	Maintenance Review Board
MTP	Maintenance Task Packaging
NP	Non-deterministic polynomial time
OEM	Original Equipment Manufacture
RCM	Reliability-Centered Maintenance
SB	Service Bulletins

Chapter 1
Introduction

1.1. Background

Ensuring the continued airworthiness and safety of commercial aircraft is the primary objective of aircraft maintenance engineering [1]–[4]. Commercial aviation operates under strict regulatory oversight imposed by authorities such as the Federal Aviation Administration and the European Union Aviation Safety Agency. These authorities require aircraft operators to establish and comply with an approved maintenance programme that ensures continued airworthiness throughout the entire service life of the aircraft [1]–[3].

In operational environments, maintenance tasks are carried out through structured maintenance events such as Line maintenance, A-checks, and C-checks [2], [5], [6]. The process of **maintenance task packaging** refers to the process of allocating individual MPD tasks into these maintenance events while respecting technical feasibility, regulatory compliance, and operational constraints [2], [3]. This activity plays a critical role in balancing aircraft availability, hangar utilization, manpower deployment, and maintenance cost [6]. The MPD specifies individual maintenance tasks and their allowable intervals based on reliability engineering principles, operational experience, and failure-prevention considerations [4], [5]. While the content of these tasks is regulator-approved and fixed, the manner in which tasks are executed and grouped in practice is subject to planning decisions [2], [3].

For this reason, maintenance task packaging represents an engineering decision-making activity rather than a purely administrative scheduling process. Each packaging decision involves explicit trade-offs between reducing repeated access and overhead costs through task consolidation and limiting aircraft downtime and peak resource demand by distributing tasks across multiple events. Evaluating these trade-offs consistently is challenging using manual planning approaches, particularly as fleet size, task volume, and planning horizon increase.

Optimization-based approaches provide a natural framework to support maintenance task packaging decisions by explicitly modelling constraints, objectives, and trade-offs. However, translating real-world maintenance planning requirements into implementable computational models remains non-trivial. Bridging the gap between regulatory-driven maintenance requirements and practical optimization models tailored to aircraft maintenance task packaging forms the central motivation of this research.

1.2. Problem Statement

Manual maintenance planning practices frequently rely on planner experience and rule-based or heuristic approaches. While such methods can produce feasible maintenance schedules, they do not systematically guarantee cost efficiency or robustness, particularly when faced with complex task interactions and extended planning horizons. Disruptive events such as the COVID-19 pandemic further exacerbated these challenges, forcing airlines to rapidly adapt maintenance activities under conditions of unpredictable fleet groundings, extended aircraft storage, and evolving regulatory requirements [7]. This situation highlighted the need for more structured and adaptive task packaging approaches that existing manual processes or general heuristics cannot consistently provide.

This study is a continuation of previous research on aircraft maintenance task packaging conducted by Marvella [8], which addressed the problem using enumeration and brute-force approaches. While that work provided useful insight into task interactions, its formulation was limited in scalability and did not guarantee optimal task packaging solutions under realistic operational constraints. Nevertheless, the conceptual structure developed in the prior study remains valuable and can be further refined. In this research, it is extended and integrated into a mathematical optimization framework and also serves as the basis for a heuristic baseline model used in comparative benchmark analysis.

Despite growing interest in optimization for maintenance planning, a clear gap remains in the literature concerning validated Mixed-Integer Linear Programming (MILP) formulations that explicitly model indivisible, task-level constraints derived from Maintenance Planning Documents (MPDs) while remaining implementable in practical planning environments. Addressing this gap—by developing and applying a tailored MILP-based task packaging model that respects real-world maintenance constraints—constitutes the core problem investigated in this research.

1.3. Research Objectives and Questions

The primary objective of this research is to develop and validate a high-fidelity Mixed-Integer Linear Programming (MILP) model for single-aircraft maintenance task packaging in order to optimize operational efficiency while guaranteeing compliance with airworthiness requirements.

The specific research objectives are:

1. To design an MILP model that precisely captures the discrete nature of maintenance tasks and relevant operational constraints.

2. To minimize the total operational cost associated with aircraft maintenance task packaging.
3. To validate the proposed model using a real-world case study from a specific maintenance, repair, and overhaul (MRO) facility.

This study seeks to answer the following research questions:

1. Can a single-aircraft MILP model guarantee an optimal maintenance task packaging solution while maintaining full compliance with all airworthiness limits for a specific case study?
2. How do strategically designed cost parameters within the MILP formulation influence the resulting maintenance task packaging and associated operational trade-offs?

1.4. Scope and Limitations

The scope of this research is intentionally constrained to manage computational complexity and ensure the validity and interpretability of the results:

- **Single-Aircraft Focus**: The proposed model is developed and validated using operational data from a single Airbus A320 aircraft within a specific fleet. Multi-aircraft interactions and fleet-level maintenance scheduling are not considered.
- **Specific Case Study Facility**: All model parameters are derived exclusively from historical data sourced from a single maintenance, repair, and overhaul (MRO) facility.
- **Defined Planning Horizon**: The analysis is limited to a 3 year planning horizon, selected to balance practical relevance with computational tractability.

While the model incorporates real-world operational data to enhance practical relevance, several limitations are acknowledged in its current form:

- **Simplified Resource Representation**: Manpower capacity is modeled in aggregate as total man-hours per maintenance package. The dynamic assignment of mechanic skill sets, certifications, and specialized tools is not explicitly represented.
- **Deterministic Planning Assumptions**: All input parameters, including task man-hours, maintenance intervals, and utilization forecasts, are assumed to be deterministic. Stochastic variations and operational uncertainties are not considered.
- **Fixed Cost Structure**: Cost parameters are treated as fixed values derived from historical averages and do not account for short-term fluctuations or non-linear cost effects.

1.5. Methodology

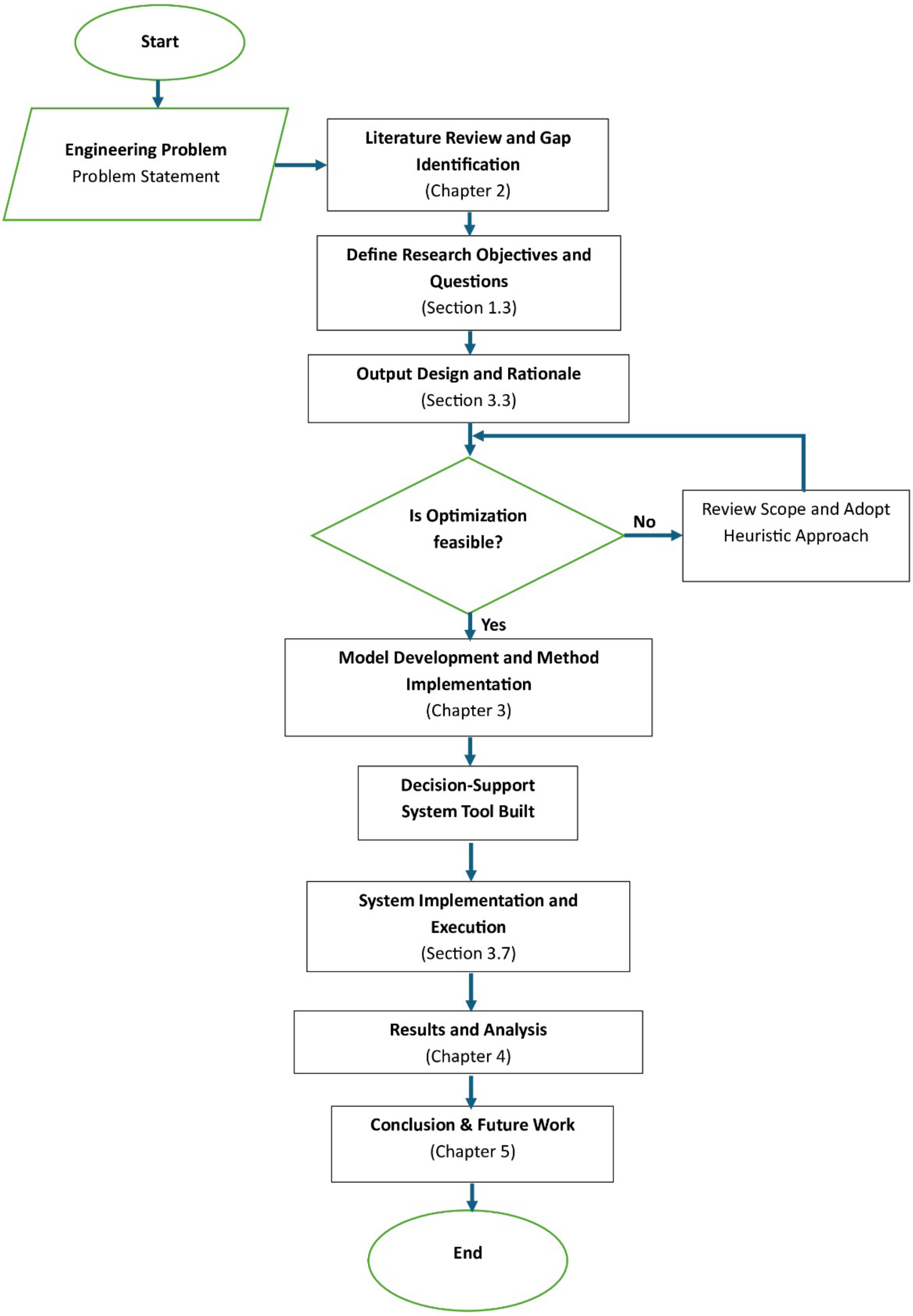

Figure 1.1 Overall Research Methodology Flowchart

This research employs an outcome-oriented design philosophy based on an optimization-driven case study approach. The primary focus is on producing implementable and interpretable maintenance task packaging solutions that reflect real-world operational constraints rather than on theoretical model development alone. Figure 1.1 presents a high-level overview of the research methodology.

The methodology begins with a comprehensive literature review to establish the theoretical and practical foundations relevant to aircraft maintenance task packaging and optimization methods (Chapter 2). This is followed by the development of a Mixed-Integer Linear Programming (MILP) model specifically tailored to the identified engineering problem, including the model assumptions, the system operation logic and procedures (Chapter 3).

The proposed model is subsequently implemented through custom-developed software and applied to operational data from a specific maintenance, repair, and overhaul (MRO) facility to conduct a detailed case study. The computational results are analyzed to evaluate model performance and practical implications for maintenance task packaging (Chapter 4). Finally, conclusions and recommendations for future research are drawn based on the findings (Chapter 5).

1.6. Report Structure

The remainder of this thesis is structured as follows:

Chapter 1 introduces the research context and motivation. It presents the background of aircraft maintenance task packaging, formulates the research problem, defines the research objectives and questions, establishes the scope and limitations of the study, and outlines the overall research methodology and report structure.

Chapter 2 presents a comprehensive review of the engineering background and relevant literature related to aircraft maintenance planning and task packaging. This chapter discusses traditional maintenance philosophies, regulatory contexts, and prior optimization-based approaches, with emphasis on methods applicable to maintenance task packaging problems.

Chapter 3 describes the development and implementation of the proposed Mixed-Integer Linear Programming (MILP) model. This chapter details the problem formulation, objective function, constraints, and the overall system operational procedure designed to reflect practical aircraft maintenance task packaging activities.

Chapter 4 presents the case study application and analysis of results. This chapter describes the application strategy, system architecture, and computational experiments conducted using data from the selected maintenance environment, followed by a discussion of the optimization outcomes and their operational implications.

Chapter 5 concludes the thesis by summarizing the main findings, addressing the research questions, and outlining the limitations of the study and potential directions for future research.

Chapter 2
Literature Review

To position this research effectively within existing engineering practices and justify the chosen methodology, this chapter provides a comprehensive review of foundational literature on airworthiness regulation, reliability-centered maintenance (RCM), economic drivers in MRO, and the operational research methods applied to aircraft task packaging. This review is crucial for demonstrating the validity of model's engineering inputs and supports the methodological choices, ensuring the research is relevant, necessary, and built upon a robust academic foundation.

2.1. Engineering Context: Airworthiness, Reliability, and Regulatory Frameworks

Aviation maintenance is fundamentally governed by the imperative of safety and continued airworthiness. Regulatory authorities such as FAA and EASA mandate strict adherence to maintenance programs derived from rigorous engineering analysis [2], [3], [9]. The foundation for commercial aircraft maintenance program is typically derived from the Maintenance Review Board (MRB) [1] process and the resulting Maintenance Planning Document (MPD) [10].

Literature confirms that reliability engineers utilize methods like Failure Model and Effect Analysis (FMEA) and Reliability-Centered Maintenance (RCM) to define critical tasks and their maximum intervals [4], [11]. These intervals are engineering-derived limits designed to mitigate identified failure risks and are legally binding. For example, damage tolerance analysis for primary structural components mandates specific inspection intervals in the Airworthiness Limitations Section (ALS) to prevent catastrophic failure from crack growth [2], [4]. The primary objective of the engineering context is compliance; the optimization problem addressed in this report is fundamentally constrained by these non-negotiable safety and regulatory mandates.

2.2. Maintenance Locations and Traditional Maintenance Philosophies

Aircraft maintenance traditionally employs two primary philosophies for packaging tasks: the block method and equalized method. These approaches define how maintenance activities are grouped, packaged and executed, directly impacting operational efficiency and compliance within the engineering field. These philosophies are also intrinsically linked to the physical location of maintenance: line (apron) and base (hangar).

2.2.1. Block Method

The **block method**, also known as the letter check system, is a traditional maintenance philosophy where tasks are grouped into predefined checks based on fixed multiples of operating time [5], [6].

- **Description:** For example, an A-check might be scheduled every 500 flight hours (FH), and a C-check every 4,000 FH. Any task with an interval shorter than or equal to the check interval is bundled into that corresponding block. This creates a pyramid structure of maintenance tasks that build upon each other.
- **Advantages:** This method is simple, easy to plan, document, and audit for regulatory compliance. Its deterministic nature aligns well with regulatory oversight.
- **Disadvantages:** It often leads to the premature execution of tasks, consuming unnecessary labor and shortening component on-wing time (e.g., a 1,000 FH task is performed twice within 2,000 FH if the block interval is 500 FH). This rigidity can be inefficient as aircraft utilization patterns evolve, they offer limited flexibility and cost efficiency.

2.2.2. Equalized Method

In contrast, the **equalized method** seeks to distribute the total maintenance workload more evenly across a series of smaller, more frequent visits [5], [6].

- **Description:** Rather than bundling all tasks at major fixed milestones, tasks are phased so that workload and access actions are balanced across the entire maintenance cycle. This approach aims to minimize peaks and valleys in manpower requirements.
- **Advantages:** This method allows the aircraft to remain operationally available for longer periods and reduces the magnitude of individual check events and downtime peaks. It offers greater adaptability to changing operational requirements.
- **Disadvantages:** The complexity of balancing thousands of interval-based tasks with multiple interdependencies renders purely manual or heuristic approaches inadequate for guaranteeing an optimal solution. It also increases the complexity of planning and management.

2.2.3. Line and Base Maintenance Context

Task packaging must also respect the organizational and physical boundaries between two primary types of maintenance locations, which influence resource availability and cost:

- **Line maintenance**: This covers frequent, low-duration tasks (e.g., transit checks, A-checks, minor repairs) typically performed between flights or overnight at the airport gate or apron. It emphasizes rapid turnaround with minimal downtime and is limited by available ground support equipment. EASA regulations define it as maintenance that can be performed outside of a hangar [2], [5], [12].

- **Base maintenance**: This involves more extensive work, such as scheduled heavy checks (C-checks, D-checks), major structural repairs, and complete overhauls that require the aircraft to be removed from service for a longer period (days to months). This work is conducted in a hangar environment equipped with specialized tools and facilities [2], [5].

The packaging problem must ensure that tasks assigned to each package are compatible with available maintenance resources and locations. While this constraint is usually treated operationally rather than mathematically, it influences packaging feasibility—certain access actions or structural inspections can only be performed under base-maintenance conditions.

The distinction between line and base maintenance is thus an implicit boundary condition in any packaging optimization model: each task's attributes (duration, access zone, equipment need) indirectly determine which maintenance environment it can belong to. This ensures the resulting packages are not only compliant and efficient but also operationally executable.

2.3. Economic Drivers of MRO and Total Cost Modelling

Optimization of maintenance operation is extremely important for airlines seeking to stay competitive, as maintenance, repair and overhaul (MRO) activities represent a significant percentage of airline's total operational costs [12], [13]. Literature confirms that effective decision-making tools must consider a holistic view of the economic drivers beyond simple labor costs [13], [14]. The total cost associated with maintenance packaging in real-world scenario involves several key components recognized in the literature:

- **Direct Maintenance Cost (DMC):** These involve fundamental expenditures on labor (technician hours, labor rates) and materials (spare parts, consumable). Literature consistently models these as primary cost drivers that fluctuate with specific task requirements [14], [15].
- **Indirect Cost and Overhead:** These encompass a wider range of expenses, including the use of hangar space, specialized tools, quality control, administration and transport (towering) aircraft to hangar. These are often complex to allocate but are essential for an accurate total cost picture [12], [13], [16].
- **Downtime and Opportunity Cost:** A significant economic impact arises from aircraft being out of service (downtime). This represents a loss of potential revenue for the airline.

Research highlights that minimizing the downtime hours is crucial because it is substantial cost driver in itself [14]–[16].

- **Penalty Cost (Behavioural):** The literature justifies the use of penalty functions within optimization models to encourage or penalize specific packaging assignment behaviours, such as executing task too early to be assign to only available package or a task is evaluated better to be execute in individually rather than in package, which helps guide the model toward the most efficient operational strategy for tasks packaging arrangement while remain compliant and reach global optimal solution for total cost as objective [15]–[17], [24].

While the literature acknowledges these varied costs, many existing optimization models tend to simplify the total cost function to manage computational complexity, often focusing only on direct maintenance cost, indirect cost or downtime cost model [17], [18]. This study addresses this limitation by synthesizing these factors into a comprehensive total cost function, as detailed in Chapter 3.

2.4. Mathematical Characterization of Maintenance Packaging Problem

The integration of these complex economic factors into a solvable framework requires a formal mathematical characterization of the problem. From an operations research perspective, aircraft maintenance task packaging is widely recognized as a complex combinatorial optimization problem [17], [19]. The decision of which indivisible task to place in which available package, while satisfying all constraints, is analogous to the classic bin-packing problem.

Packaging consolidates tasks that share compatible intervals or access requirements so that multiple inspections or replacements can be done concurrently. This minimizes repeated disassembly and reassembly, reduces aircraft downtime, and optimizes manpower utilization. A well-designed package aligns with both regulatory compliance and operational economics: all tasks remain within their limits, and total downtime over a maintenance cycle is minimized.

Each package can therefore be viewed as a *logical container* of tasks. For a given MPD with thousands of tasks, determining which task belongs in which package becomes a **combinatorial optimization** problem [17]. This problem grows rapidly in complexity as interval combinations, access overlaps, and cost interdependencies multiply. The following philosophies historically emerged to manage this complexity manually or heuristically.

Research has established that solving this problem for a large fleet in dynamic environment is NP-hard [17], [20]. This classification is significant because it explains why findings a mathematically proven optimal solution for the entire industry is computationally prohibitive within practical timeframes. The literature justifies the engineering trade-off between seeking a perfect solution versus a timely, feasible one, which informs the scope of this research.

2.5. Review of Solution Methodologies

Due to the NP-hard nature of large-scale maintenance task packaging, the literature presents a variety solution methodologies, each with distinct advantages and drawbacks :

- **Heuristics and Metaheuristics:** Studies frequently employ genetic algorithms (GA), simulated annealing, or custom heuristics to solve large, fleet-wide packaging problems quickly. These methods provide good, near optimal solutions but often lack mathematical guarantee of global optimality for verifiable compliance in safety-critical system [11], [18], [22].
- **Mixed-Integer Linear Programming (MILP):** Research also utilizes MILP for maintenance task packaging, particularly where precision and optimality are critical. MILP model excel at handling discrete decisions (binary variables) and complex constraints, guaranteeing an optimal solution if computational limits are respected. The trade-off is often the time required to solve large-scale problems [21], [23].
- **Hybrid Frameworks:** Emerging literature suggests combining methodologies to leverage strengths: using heuristics for high-level fleet management (macro-decisions) and optimization solvers like MILP for tactical, task-level packing problems (micro-decisions) [16], [17].

Existing research on aircraft maintenance optimization spans a wide range of methodological approaches, including exact mathematical programming, heuristics and metaheuristics, data-driven algorithms, and hybrid frameworks. These approaches differ substantially in terms of problem scope, cost representation, regulatory fidelity, and decision granularity. Some studies focus primarily on task allocation or resource scheduling, while others emphasize facility layout, fleet-level coordination, or predictive maintenance strategies.

To systematically assess the extent to which existing literature addresses the aircraft maintenance task packaging problem under realistic operational and economic constraints, a comparative review is conducted. The reviewed studies are evaluated along key dimensions, including the underlying optimization methodology, modeling scope, representation of maintenance costs and downtime, and alignment with Maintenance Planning Document (MPD)–driven task structures. The results of this comparative assessment are summarized in Table 2.1.

Table 2.1 Existing Studies on Methodologies Solution

Reference	Methodology Used	Problem Scope	Cost Formulation	Key Strength	Key Limitation
Ali, Rudinskas & Leonavičiūtė (2025) [21]	Mixed-Integer Linear Programming (MILP)	Small fleet maintenance task allocation (Cessna 172 case study)	Simplified Operational Cost	Explicitly models manpower skill heterogeneity; Rigorous MILP formulation with exact optimization guarantees	Limited to small fleets; MATLAB prototype, focus is on allocation rather than long-horizon packaging
Witteman, Deng & Santos (2021) [17]	Constructive heuristic (bin-packing, worst-fit decreasing)	Multi-year fleet maintenance task allocation	Indirect cost via bin utilization	Computationally efficient and scalable bin-packing formulation	Sacrifices global optimality; heuristic tuning required, Abstracted from regulatory constraints
Filho et al. (2024) [16]	Exact algorithm + Integer Linear Programming (ILP); Optimization-based (MILP/heuristic hybrid)	Task allocation and packing (MSG-3 based requirements)	Penalty-based cost formulation	Incorporates task relationships, preparation tasks, and opportunity costs. (Explicit consideration of feasibility and packing penalties)	Computationally heavy; requires detailed reliability/failure data
Andrade et al. (2021) [22]	Deep Q-learning (Reinforcement Learning)	Long-term fleet check scheduling (A- and C-checks)	Implicit reward-based cost	Adaptive and data-driven scheduling approach; reduces number of checks; robust to disturbances	Requires large training datasets; constraint satisfaction depends on training data and reward design

Reference	Methodology Used	Problem Scope	Cost Formulation	Key Strength	Key Limitation
Pazhooh & Shemirani (2025) [23]	Continuous-Time MILP	Integrated hangar scheduling and layout optimization	Detailed Scheduling Cost	High-fidelity continuous-time modeling; exact optimization feasible	Not MPD-driven; focuses on system- and facility-level scheduling rather than task packaging.
Tseremoglou, Scarpin & Bertolini (2023) [11]	Comparative modeling (MILP, heuristics, metaheuristics)	Condition-based fleet maintenance scheduling	Varies by model	Benchmarks multiple approaches; highlights trade-offs	Review-oriented; not a deployable packaging or scheduling model
Vu et al. (2024) [18]	Genetic Algorithm (GA)	Predictive maintenance optimization in industrial systems	Maintenance-related cost function	Demonstrates GA's adaptability for predictive maintenance; scalable to large datasets; Effective for complex nonlinear search spaces	Conceptual rather than aviation-regulatory specific; requires adaptation to aircraft maintenance context

As shown in Table 2.1, existing studies tend to emphasize either computational tractability, system-level integration, or methodological novelty, often at the expense of economic fidelity or regulatory realism. Heuristic and metaheuristic approaches demonstrate scalability and flexibility but generally lack guarantees of feasibility under strict MPD constraints. Conversely, exact MILP-based formulations offer mathematical rigor but frequently simplify cost structures or omit explicit modeling of aircraft downtime and opportunity loss.

Moreover, several studies address scheduling or maintenance decision-making at the fleet or facility level, rather than focusing on the detailed packaging of individual MPD-defined tasks for a single aircraft. As a result, the interaction between task intervals, packaging feasibility, and cost accumulation remains insufficiently captured. This observation highlights a persistent research gap for an optimization framework that integrates MPD-driven task packaging with explicit cost representation and operational feasibility at the single-aircraft level.

2.6. Synthesis and Research Gap

The existing study of literature provides a strong foundation for both the engineering requirements and the optimization techniques needed for aircraft maintenance task packaging. However, many existing MILP studies tend to oversimplify the operational realities and task-level constraints, while most heuristic models lack the rigorous mathematical guarantee of full compliance required for airworthiness verification [16], [17], [11].

Modern optimization models, such as the one presented in this study, aim to numerically balance traditional philosophies and maintenance location classification by incorporating specific cost parameters that capture the economic impact of both downtime and fixed overheads (e.g., in this research model's 'overhead cost' and 'downtime cost' component which will be explain in next Chapter 3), thus finding a new equilibrium for decision-making.

This study also aims to bridge the gap in many existing MILP studies by developing a high-fidelity MILP model focused on the single-aircraft task packaging problem. By achieving a mathematically optimal solution for this core problem scope, this research provides a rigorous, validated baseline that can inform the design of future, more complex hybrid metaheuristic frameworks.

Guided by the limitations identified in the literature review, this study adopts a mixed-integer linear programming (MILP) framework to explicitly model the aircraft maintenance task packaging problem at the single-aircraft level. In contrast to prior work that treats maintenance activities in aggregated form, the proposed model represents each task as a discrete decision entity associated with a defined interval, execution duration, and cost contribution.

Consistent with the terminology introduced in Chapter 3, maintenance tasks are indexed by the set $i \in I$, while packaging decisions are represented through binary decision variables that assign tasks to specific maintenance opportunities or checks. Cost components are explicitly structured to distinguish between direct maintenance cost and downtime-related effects, enabling a transparent mapping between operational decisions and economic outcomes. By embedding these elements within a unified MILP formulation, the proposed approach directly addresses the regulatory, economic, and combinatorial characteristic highlighted as gaps in the existing literature.

Chapter 3
Model Formulation and MILP Implementation

Building upon the literature-derived research gap established in Chapter 2, the following chapter formalizes the aircraft maintenance task packaging problem using a MILP model, defining sets, parameters, decision variables, and assumptions accordingly. This chapter details the development of the Mixed-Integer Linear Programming (MILP) model used to solve the single-aircraft maintenance task packaging problem. This section moves from theoretical background to the precise application of optimization techniques. The model is uniquely design with an outcome-oriented philosophy, ensuring the resulting packaging is both mathematically optimal and a practical, verifiable engineering document.

3.1. Problem Detail, Formulation Model and Experimental Assumptions

To manage the NP-hard nature of the task packaging problem (Section 2.4) while maintaining rigor, this study focuses on a simplified, single-aircraft scope (Section 1.4). The core challenge is packaging indivisible tasks into a sequence of available maintenance packages to minimize total cost. The dynamic nature of real-world operations, such as fluid manpower shifts and real-time downtime calculation, is addressed using simplified linear assumptions to ensure the model remains solvable.

The model uses the following key parameters and assumptions derived from the case study data:

3.1.1. Notation, Set, Parameters and Decision Variables

- **Indices and Set**
  - **Aircraft Planning Horizon:** H
  - **Tasks:** I indexes tasks $i \in I$
  - **A-interval candidates:** A indexes discrete A-check candidates $a \in A$
  - **C-interval candidates:** C indexes discrete C-check candidates $c \in C$
  - **Candidate bins:** T indexes bin endpoints $t \in T$ where:

$$T = \{t \in \{1, \dots, H\} \mid t \text{ is multiple of some } a \in A \text{ or } c \in C\}$$

  - **Allowed assignment domain:** $T_i = \{t \in T | t \leq I_i\}$
- **Parameters:**
  - MH_i: Man-hours required for Task i (deterministic value from total manhours component)

- I_i: Minimum Interval available for Task i (in flight hours, a deterministic minimum value from all interval value of a task after normalization with average value of utilization forecast data to represent maximum dues for when the task must be executed before due date)
- M_{cap}: Manpower capacity per shift of Package t

- **Decision Variables**
  - **A/C interval selection:**

$$s_a^A \in \{0,1\} \ \forall a \in A, \qquad with \sum_{a \in A} s_a^A = 1$$

$$s_c^C \in \{0,1\} \ \forall c \in C, \qquad with \sum_{c \in C} s_c^C = 1$$

  - **Bin opening and assignment:**

$$v_t \in \{0,1\}, \qquad \forall t \in T(bin\ open\ indicator)$$

$$x_{i,t} \in \{0,1\}, \qquad \forall i \in I, \forall t \in T_i\ (task\ i\ assign\ to\ bin\ t)$$

  - **Line decision:**

$$y_i^{Line} \in \{0,1\}\ (1\ if\ task\ i\ left\ to\ line; only\ allowed\ for\ cat_i = 1)$$

  - **Classification variables**

$$isA_t \in \{0,1\}\ and\ isC_t \in \{0,1\}\ (bin\ type\ indicators)$$

$$z_t^A \in \{0,1\}\ and\ z_t^C \in \{0,1\}\ (overhead\ trigger)$$

3.1.2. Assumption and Simplifications

- **Assumptions and Simplifications:**
  - **Single-Aircraft:** The study focuses on one aircraft type — **A320 family** — as a representative narrow-body platform. Only tasks with defined intervals (FH, FC, and calendar) are considered
  - **Downtime is a Resource Constraint**: The complex calculation of aircraft downtime hours is linearly modelled as a fixed resources using formula that approach its output result when compare to real-world dynamic execution (**Appendix A)** [15], [16]. Integration of above parameter to the simplified formula will help the simplified formula to better representing the dynamic model shift execution complexity.
  - **Fixed Manpower:** Labor is aggregated into the M_{cap} parameter per shift and the C_{Labor} cost, ignoring dynamic skill sets for tractability

- **Deterministic Environment:** All data inputs are assumed to be known and fixed during the optimization run.

Table 3.1 Symbol and Parameters for MILP-based Model Formulation

Symbol	**Description**
I_i	Task Interval
MH_i	Task Man-hours
ϕ	Material Cost Factor
φ	Labor Rates
ω	Overhead Cost per Events
ρ	Revenue Rate per hours
I_t	Bin Assignment Interval
I_A , I_C	Interval A Check and C Check chosen
d_T	Downtime Hours
$e_{i,t}$	Early Execution Penalty (in Flight Hours)
y_i^{Line}	Line Task
M_i, M_t	Manpower availability for task i and package t respectively
H	Observed Planning Horizon

3.2. Objective Function Formulation

The primary objective of this research model is to determine the optimal packaging arrangement of maintenance tasks for a single aircraft to minimize the total associated costs within the defined planning horizon, while adhering to all regulatory and operational constraints. As established in Chapter 2, existing literature often simplifies cost structures to manage computational complexity. The objective function derived for this study is a comprehensive synthesis design to capture all major direct and indirect economic drivers, including labor, material, overhead, and aircraft downtime.

The optimization model balances compliance with airworthiness directives against efficient resources utilization, operational management, and the value of maximizing task intervals. Importantly each cost components formula will be adjusted to MILP-based system but still respect to how real-world execution of maintenance tasks are. The total cost function (Z_{total}) is defined as the summation of direct maintenance costs (DMC), indirect maintenance costs (IMC), and economic penalties, as formulated in Equation 3.1 to ensures the optimal packaging assignment is the most cost-efficient while adhering to all constraints.

$$Minimize\ (Z_{total}) = C_{Labor} + C_{Material} + C_{Overhead} + C_{Downtime} + C_{Penalty} + C_{Line} \quad 3.1$$

The subsequent sections detail each cost component, its operational relevance, and its implementation within Mixed-Integer Linear Programming (MILP) model.

3.2.1. Labor Cost (C_{Labor}) and Material Cost ($C_{Material}$)

- **Rationale:** These components represent direct operating cost component and the primary variable operating costs of maintenance execution. Accurate modelling of labor rates and material consumption using factor estimation (ϕ) is essential for comparing the financial viability of different packaging strategies. Material cost factor (ϕ) value represents relation between labor and material cost assumptions and estimation.
- **Formulation:** These are derived from how maintenance task executed and assumption as mention, calculated as sums of the resources consumed. These formulas will be calculated for each task:

$$C_{Labor} = \sum_{i \in I} \sum_{t \in T} \varphi \cdot MH_i \cdot x_{i,t} \quad 3.2$$

$$C_{Material} = \phi \cdot C_{Labor} \quad 3.3$$

Where $x_{i,t}$ is the binary decision variable assigning task i to available bin/package t.

3.2.2. Overhead Cost ($C_{Overhead}$)

- **Rationale:** Overhead captures the indirect operating costs associated with utilizing maintenance facilities and infrastructure as detail in Table 3.1. This cost incentivizes shorter maintenance durations and efficient utilization of high-value maintenance bases.

(Note: Insert Table 3.1: List of activities include in Overhead Cost (e.g., hangar space, utilities, aircraft tow)

This cost components also captures of how different treatment regarding tasks with special environment requirement (hangar facilities) as mention in Section 2.2.3. Base maintenance tasks will be mark later on in preprocessing process and explain in detail at Section 3.5. As this cost component calculated for each maintenance events, if in the event no task needs hangar arrangement then the event will be executed in Apron and Overhead Cost will be calculated accordingly. This formulation is accordance to objective of minimize total cost and respect real-world execution situation.

- **Formulation:** Calculated based on the list on Table 3.1 that happen for each maintenance event and respecting how real-world situation work:

$$C_{Overhead} = \sum_{t} \omega_{Apron} \cdot z_t^A + \sum_{t} \omega_{Hangar} \cdot z_t^C \qquad 3.4$$

3.2.3. Downtime Cost ($C_{Downtime}$)

- **Rationale:** The downtime cost is the primary cost component that signify loss of revenue of grounding an aircraft. It quantifies the lost of revenue that the aircraft would have generated if it were operational. This metric ensures that the economic impact of grounding the aircraft is fully internalized by the optimization model.
  The grounding time will be calculated using simplified linear model from its detail formulation explain in **Appendix A** (its detail execution calculation and justification approach). To respect how real-world execution of downtime calculation, approximation linear formula is designed that tailor with MILP-based model formulation. This design that represented a shift manpower and resources model produce result with approximate values close to how real-world execution of downtime hours calculated (explain in **Appendix A**).
- **Formulation:** The core challenge lies in accurately estimating the duration of the maintenance package for integration into the MILP model. This study employs a simplified linear formula which utilizes engineering principle to provide robust estimation of total maintenance downtime (d_T). Calculated as the revenue rate per hour multiplied by the total maintenance downtime (duration of the grounding event):

$$C_{Downtime} = \sum_{t \in T} \rho \cdot d_T \quad 3.5$$

$$d_T = \frac{MH_t}{U \cdot E \cdot M_t} \quad 3.6$$

- **Validation:** Modelling the granular, dynamic execution of maintenance tasks (where manpower is continually reallocated) is non-linear and computationally intensive, incompatible with efficient MILP optimization. The linear model simplifies this by focusing on strategic resource planning rather than minute-by-minute execution tracking. This model validated against a detail Dynamic Execution Model which simulates phase-by-phase reallocation of manpower.
  The detailed calculation for the Dynamic Execution Model can be found in **Appendix A**. The near-identical result justify the use of the simplified model to represent more detail execution within the MILP framework, ensuring the downtime cost component is both robust and validated.

3.2.4. Penalty Cost ($C_{Penalty}$)

- **Rationale:** This component penalizes the early execution of maintenance tasks relative to their optimal due date. This cost component is the primary opportunity cost model to the system representing trade-offs of chosen package assignment instead of its best assignment for global optimum result objective (total cost) to task packaging model formulation. In aviation, maximizing the interval between maintenance events is crucial for minimizing the life-cycle cost of the aircraft. Performing a task early consumes valuable "interval life" prematurely, leading to more frequent maintenance cycles over the aircraft's lifespan. The penalty converts this inefficiency into measurable cost, encouraging the alignment of maintenance packages with the precise regulatory limits.
- **Formulation:** The penalty cost is calculated based on the unused interval hours consumed by early execution. This promotes optimal interval management, where a task is performed 100 flight hours prior to its 500 flight hours limit incurs a penalty proportional to the lost operational time:

$$C_{Penalty} = \rho \cdot \sum_{i \in I} \sum_{t \in T_i} e_{i,t} \cdot x_{i,t} \cdot \left(\frac{H}{I_t}\right) \quad 3.7$$

$$e_{i,t} = I_i - I_t \quad 3.8$$

This formulation ensures that solutions aligning execution times closer to regulatory limits are favored economically.

3.2.5. Line Cost (C_{Line})

- **Rationale:** The line cost serves as an alternative cost mechanism when a task cannot be efficiently packaged into a maintenance package. It represent the cost executing a task individually (e.g., as a quick "overnight check" between flights) outside of a major maintenance package. This flexibility allows the optimizer to compare the benefits of packaging versus individual execution.
- **Formulation:** Activated by a binary variable $y_L[i]$ if task i is assigned individual execution:

$$C_{Line} = \sum_{i \in I} y_i^{Line} \cdot \left(\frac{H}{I_i}\right) \cdot \left((1+\phi) \cdot \varphi \cdot MH_i + \frac{MH_i}{M_i} \cdot \rho + \omega_{Apron}\right) \qquad 3.9$$

3.2.6. Interaction and Model Behavior

Based on all cost components definition and how its mathematical formula, the final form of total cost and objective of this research formulation model is expressed as follow:

$$\begin{aligned} Z = & \underbrace{(1+\phi) \cdot \sum_{i \in I} \sum_{t \in T} \varphi \cdot MH_i \cdot x_{i,t}}_{DMC} + \underbrace{\sum_{t \in T} \left(\omega_{Apron} \cdot z_t^A + \omega_{Hangar} \cdot z_t^C\right)}_{Overhead\ Cost} \\ & + \underbrace{\rho \cdot \sum_{t \in T} \frac{MH_t}{E \cdot M_t}}_{Downtime\ Cost} + \underbrace{\rho \cdot \sum_{i \in I} \sum_{t \in T_i} e_{i,t} \cdot x_{i,t} \cdot \left(\frac{H}{I_t}\right)}_{Penalty\ Cost} \\ & + \underbrace{\sum_{i \in I} y_i^{Line} \cdot \left(\frac{H}{I_i}\right) \cdot \left((1+\phi) \cdot \varphi \cdot MH_i + \frac{MH_i}{M_i} \cdot \rho + \omega_{Apron}\right)}_{Line\ Cost} \end{aligned} \qquad 3.10$$

This mathematical formula will better represent how interplay and correlation of each cost components and its intended behavior. From interplay between these components drives the optimization behavior. High *Downtime* and *Overhead* costs incentivize packaging tasks into fewer, consolidated events. Simultaneously, the *Penalty (Opportunity)* cost prevents suboptimal "pulling forward" of tasks into the earliest possible package, pushing the solution toward maximizing utilized intervals. The *Line* cost offers a pathway to unpackage tasks if their inclusion significantly increases the duration or penalty costs of main package. This dynamic balance ensures the optimized packaging arrangement is both cost-efficient and operationally robust.

3.3. Output Design Rationale and Method Suitability

The model's design was driven by engineering utility and visualization of its output: they must be auditable, traceable, and directly usable in aircraft maintenance task packaging assignment. Outputs are structured to provide discrete assignment decisions, clear visualization, and cost decomposition that can be verified against regulatory requirements. This section explains why outputs are designed in this way, how they justify the use of Mixed-Integer Linear Programming (MILP), and how they correlate with the broader model formulation.

3.3.1. Discrete vs. Continuous Results

Aircraft maintenance programs operate under **discrete regulatory intervals** (e.g., A-checks at 600 FH, C-checks at 6000 FH). Continuous optimization of check timing would yield fractional or non-divisible results that cannot be executed in practice. The model therefore produces **binary assignment decisions**:

$$x_{i,t} \in \{0,1\}, \quad \text{“Task } i \text{ is assigned to bin } t\text{”.}$$

This ensures outputs answer the engineering question in precise terms: *"Does Task A go to Package 1 (Yes/No)?"* rather than producing ambiguous fractional allocations.

- **Engineering justification:** Discrete outputs align with maintenance manuals and airworthiness regulations, where checks are defined at fixed flight-hour multiples.
- **Method suitability:** MILP is the natural framework for such combinatorial decisions, as it can guarantee global optimality across discrete candidate sets.

3.3.2. A- and C-interval selection

The choice of the "best" A- and C-intervals (I_A *and* I_C respectively) are also modelled as a **discrete selection problem**. The solver must commit to exactly one $a^* \in A$ and one $c^* \in C$. This design reflects engineering reality: operators adopt fixed policies for A- and C-checks, not continuous or fractional intervals.

- **Engineering justification:** Discrete interval selection ensures that the resulting packages are consistent with actual maintenance programs and can be certified.
- **Method suitability:** MILP allows endogenous policy selection within the same optimization, ensuring that interval choice and task packaging are jointly optimal.

3.3.3. Traceability and Auditability

Outputs are designed as **matrix visualizations** (task-to-package assignment tables) and **locked schedules** (occurrence lists). These formats allow certifying engineers to verify every task's placement against regulatory requirements.

- **Engineering justification:** Maintenance planning must withstand regulatory audit; outputs must be traceable from decision variables to occurrence schedules.
- **Method suitability:** MILP's binary assignment structure translates directly into auditable tables, avoiding the opacity of heuristic or continuous approaches.

3.3.4. Influence on Methodology

The requirement for **discrete, verifiable outputs** and a **mathematically guaranteed optimal solution** led to the selection of MILP over heuristic methods. Heuristics may produce feasible packaging assignment but cannot guarantee global optimality.

- **Engineering justification:** Reliability and cost efficiency in aircraft maintenance demand solutions that can be certified and defended.
- **Method suitability:** MILP ensures that the chosen intervals, bin openings, and task assignments minimize total cost subject to operational constraints, producing outputs that are both optimal and certifiable.

3.3.5. Correlation with Model Design

Each output designed to corresponds directly to a modelling construct, it correlate to objective function (Section 3.2), Model's Constraints (Section 3.4) and Model's formulation:

- **Task-to-package matrix** ↔ indivisible packaging constraint.
- **Locked schedule** ↔ divisibility-based bin classification and occurrence propagation.
- **Package summary** ↔ capacity and workload aggregation constraints.
- **Cost decomposition** ↔ proxy vs exact reconciliation in the objective function.

This correlation ensures that outputs are not post-hoc approximations but direct reflections of the mathematical formulation, reinforcing their engineering credibility.

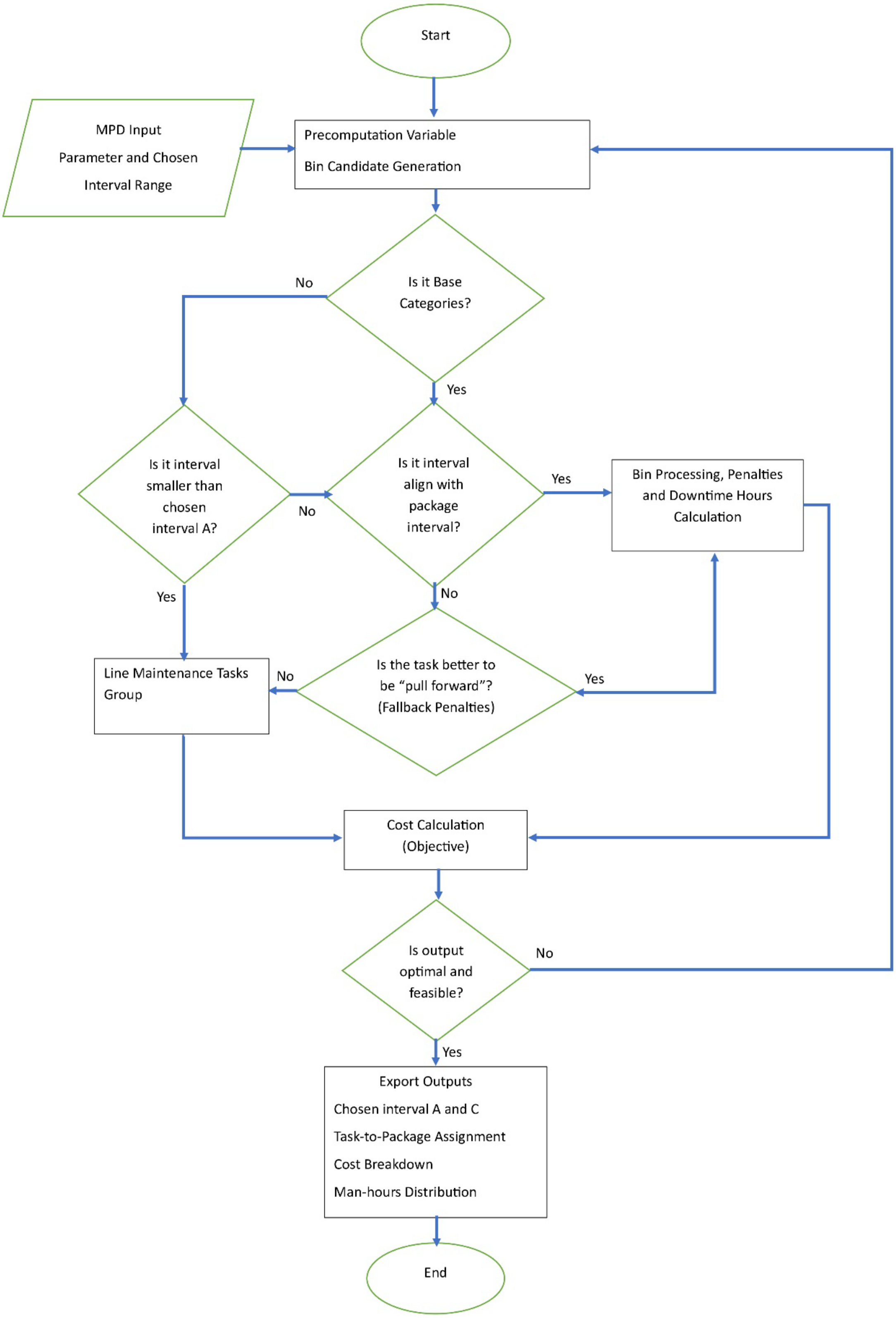

Figure 3.1 Conceptual Logic Flow of the MILP Optimization Model

3.4. Constraints for Design System (MILP-based)

The following constraints guarantee the feasibility and compliance of the maintenance task packaging execution:

3.4.1. Policy Selection

The model captures strategic interval selection by choosing exactly one A-Check and C-Check interval that best suits the task portfolio and economics over the horizon, rather than optimizing assignment under a fixed policy.

$$\sum_{a \in A} s_a^A = 1, \qquad \sum_{c \in C} s_c^C = 1 \tag{3.11}$$

This constraint forces the solver to commit to single A and C interval globally, enabling the MILP to co-optimize interval choice and packaging.

3.4.2. Bin Classification

The model keeps bin semantics consistent with chosen intervals, avoiding static pre-labeling that could misclassify bins when A-check and C-check intervals change.

$$isA_t = \sum_{a \in A} s_a^A \cdot 1\{a|t\}, \qquad isC_t = \sum_{c \in C} s_c^C \cdot 1\{c|t\}, \qquad \forall t \in T \tag{3.12}$$

This constraint classifies each bin endpoint t as A-type or C-type depending on the chosen intervals (A-check and C-check intervals exact divisibility). This keeps bin semantics consistent with maintenance policy after interval selection.

3.4.3. Overhead Trigger Linearization

The model uses z_t^A, z_t^C to avoid bilinear cost terms while retaining exact logic: overheads are charged only when a bin is both opened and of the corresponding type.

$$\begin{aligned} &z_t^A \le isA_t, \qquad z_t^A \le v_t, \qquad z_t^A \ge isA_t + v_t - 1 \\ &z_t^C \le isC_t, \qquad z_t^C \le v_t, \qquad z_t^C \ge isC_t + v_t - 1 \end{aligned} \tag{3.13}$$

This constraint enforces $z_t^A = isA_t \cdot v_t$ and $z_t^C = isC_t \cdot v_t$ without bilinear terms, so overhead costs apply only when a compatible bin is opened. This will preserves linearity and exact logic.

3.4.4. Bin Compatibility Constraints

To prevent “cheat” happen by the solver, the model design to only open compatible bins and assign task i only to compatible open bin t.

$$\begin{aligned} &v_t \leq isA_t + isC_t, \qquad \forall t \in T \\ &x_{i,t} \leq isA_t + isC_t, \qquad \forall i \in I, \qquad \forall t \in T_i \end{aligned} \tag{3.14}$$

These constraints prevent opening bins that are incompatible with the selected A/C intervals, strengthening the LP relaxation and eliminating non-sensical openings, also ensuring tasks are assigned only to valid A/C bins under the chosen intervals, preserving operational feasibility.

3.4.5. Link Assignment to opened bins

The model is design to prevent task to closed bin. This constraint coupled with bin compatibility constraints strengthen the LP relaxation and reducing branch-and-bound effort.

$$x_{i,t} \leq v_t, \qquad \forall i \in I, \qquad \forall t \in T_i \tag{3.15}$$

This constraints prohibits assignment to closed bins, coupling workload decisions with bin opening costs, aligning costs and capacity with actual task packaging assignment.

3.4.6. Packaging and Line Decisions

The model restricting line to $cat_i = 1$ enforces operational boundaries. The line proxy captures expected recurring cost over planning horizon (H), making the packaging vs line trade-off transparent and economically grounded.

$$\sum_{t \in T_i} x_{i,t} + y_i^{Line} = 1, \qquad \forall i \in I \tag{3.16}$$

This constraint ensure each task is either packaged once (anchored to a single bin $\leq I_i$) or entirely left to line. This single anchor propagates occurrences across the horizon in post-solve validation.

3.4.7. Task Due Date (Assignment Domain Adherence) Constraints

Tasks are assigned before their intervals, in other word its bin interval (bin/package where the task is assigned) must not exceed due date (task interval).

$$x_{i,t} = 0, \qquad if\ I_t > I_i \tag{3.17}$$

This constraint enforces strict compliance by construction—variables $x_{i,t}$ are only created for $I_t \leq I_i$, preventing anchors beyond permissible intervals and simplifying the model.

3.4.8. Bin Workload Aggregation

This constraint defines a linear aggregate used by the downtime proxy in the objective, maintaining convexity and capturing workload intensity.

$$MH_t = \sum_{i \in I} MH_i \cdot x_{i,t}\,, \qquad \forall t \in T \tag{3.18}$$

3.4.9. Line Eligibility Restriction

This constraint enforces operational policy that only line-eligible tasks can be left to line maintenance, preventing unrealistic deferrals. The model will assigned task with $I_i \leq I_A$ automatically treated as Line Task.

$$\begin{aligned} T_i = \emptyset \ &\Rightarrow y_t^{Line} = 1 \\ cat_i \neq 1 \ &\Rightarrow y_t^{Line} = 0 \end{aligned} \tag{3.19}$$

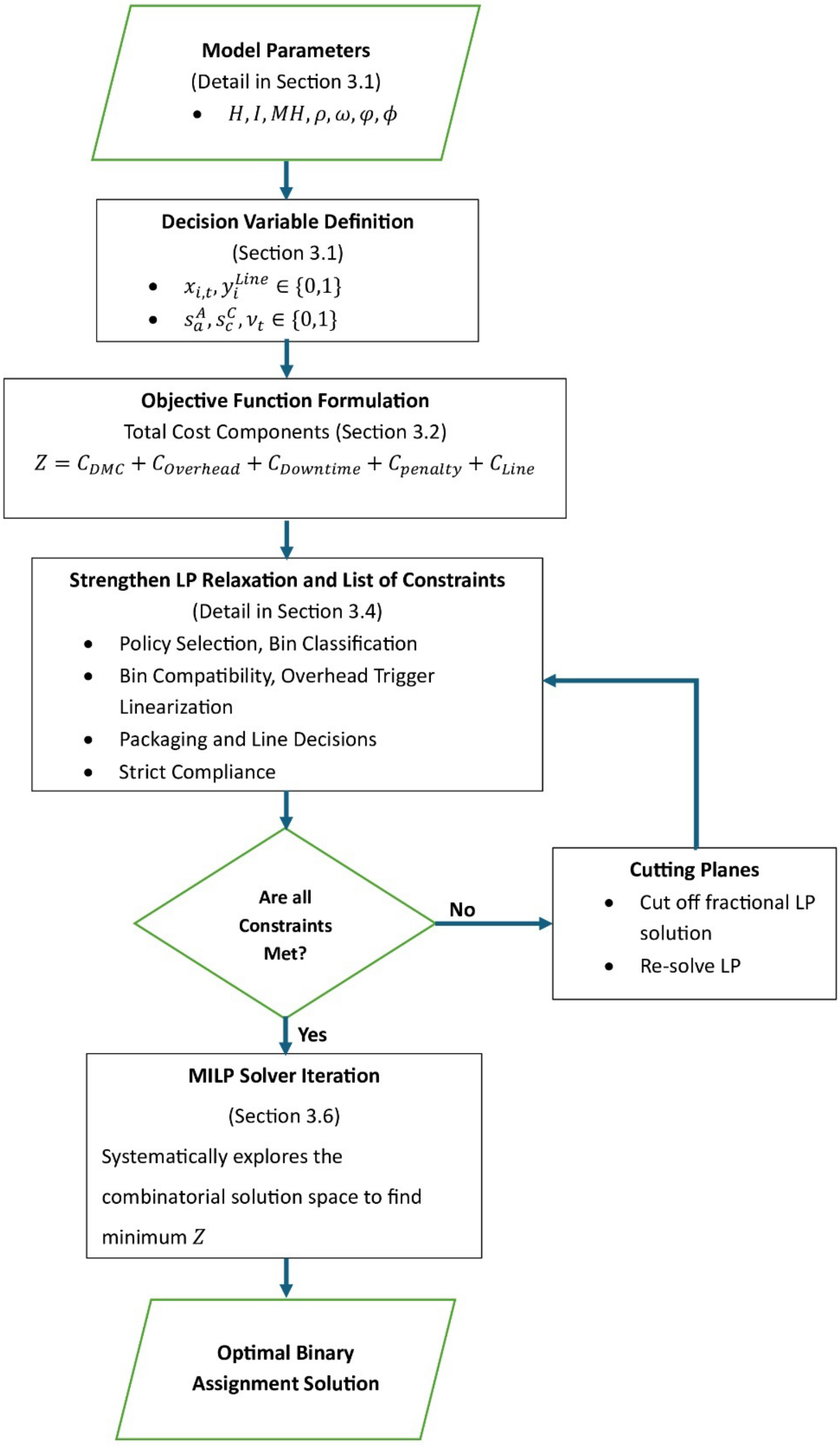

Figure 3.2 MILP Implementation on Design Decision-Support System

3.5. Data Acquisition and Preprocessing Methodology

The methodology for preparing data inputs is crucial for ensuring integrity and operational realism. The preprocessing pipeline was designed to transform raw Maintenance Planning Document (MPD) spreadsheets into normalized, auditable datasets suitable for optimization. Each step was engineered to preserve traceability, enforce engineering validation, and align with utilization assumptions defined in Section 3.1.

- **Data Sources:** Raw MPD data and historical maintenance records are sourced from the MRO's maintenance tracking system [25]. These inputs are validated by maintenance engineers to ensure engineering compliance and operational relevance.
- **Preprocessing Steps:** Data is cleaned and validated to ensure compliance with engineering standards.
  - **Column normalization:** Raw headers are mapped to canonical names (e.g., *TASK NUMBER → TaskID*, *TASK M.H. → TaskMH*) to establish a consistent schema.
  - **Interval conversion:** Heterogeneous thresholds and intervals (YE, MO, DY, FC, FH) are parsed and converted into flight-hour equivalents using utilization constants from *Utilization Forecast Data* (*FH per day, FH per cycle, FH per month*). Flags such as NOTE and CE are extracted to preserve engineering metadata. (NOTE: Refer to the note in the task description; and CE(Customizable Equipment): Consult the vendor's documentation for this interval/threshold) ([**MPD A320 FILE**]).
  - **Fallback logic:** Missing or zero values in "Sample" fields are backfilled from "100%" fields to avoid loss of interval information.
  - **Manhour aggregation:** Multiple manhour-related columns (TaskMH, AccessMH, PrepMH, Manhours) are normalized and summed into a single runtime labor field (*ManhoursRT*).
  - **Technician count:** The *Men* field is coerced to numeric, with a *MenZero* flag indicating missing or zero values.
  - **Engineering flags:** Time Controlled Items (TCI) are standardized to binary values, and Task Packaging System (TPS) indicators are parsed into *TPS1, TPS2, TPS3*.
  - **Environment eligibility:** The *Check_Categories* field is mapped to a binary *Categories* flag: Line/apron-executable tasks are encoded as 1, while Base/Hangar-only tasks are encoded as 0. This distinction is critical for downstream cost modeling.
  - **Traceability:** A sequential *RowID* is assigned to each record, ensuring reproducibility across CSV and Excel outputs.
  - **Canonical interval field:** A unified *IntervalFH* column is created, representing the normalized flight-hour interval for each task.

- **Outputs:**
    - **Cleaned CSV:** A full normalized dataset is exported for archival and audit purposes.
    - **Optimizer input subset:** A filtered dataset containing only relevant fields (*RowID, TaskID, IntervalFH, ManhoursRT, Men, Categories, Zone*) is exported to optimizer_input.csv. This ensures the optimizer operates only on validated, model-ready parameters.
    - **Excel export:** An optional formatted Excel file is generated for engineering review, with auto-width columns and styled headers for readability
- **Justification:** This rigorous preprocessing (detailed further in **Appendix A** and **Appendix B**) ensures that the mathematical model operates on reliable, engineering-approved data. By explicitly encoding environment eligibility (*Categories*), the methodology captures overhead implications: tasks executable on the apron (*Categories = 1*) minimize facility overhead, while Base/Hangar tasks (*Categories = 0*) incur additional costs. This distinction is essential for accurate cost modeling and ensures that optimization results reflect real-world execution constraints.

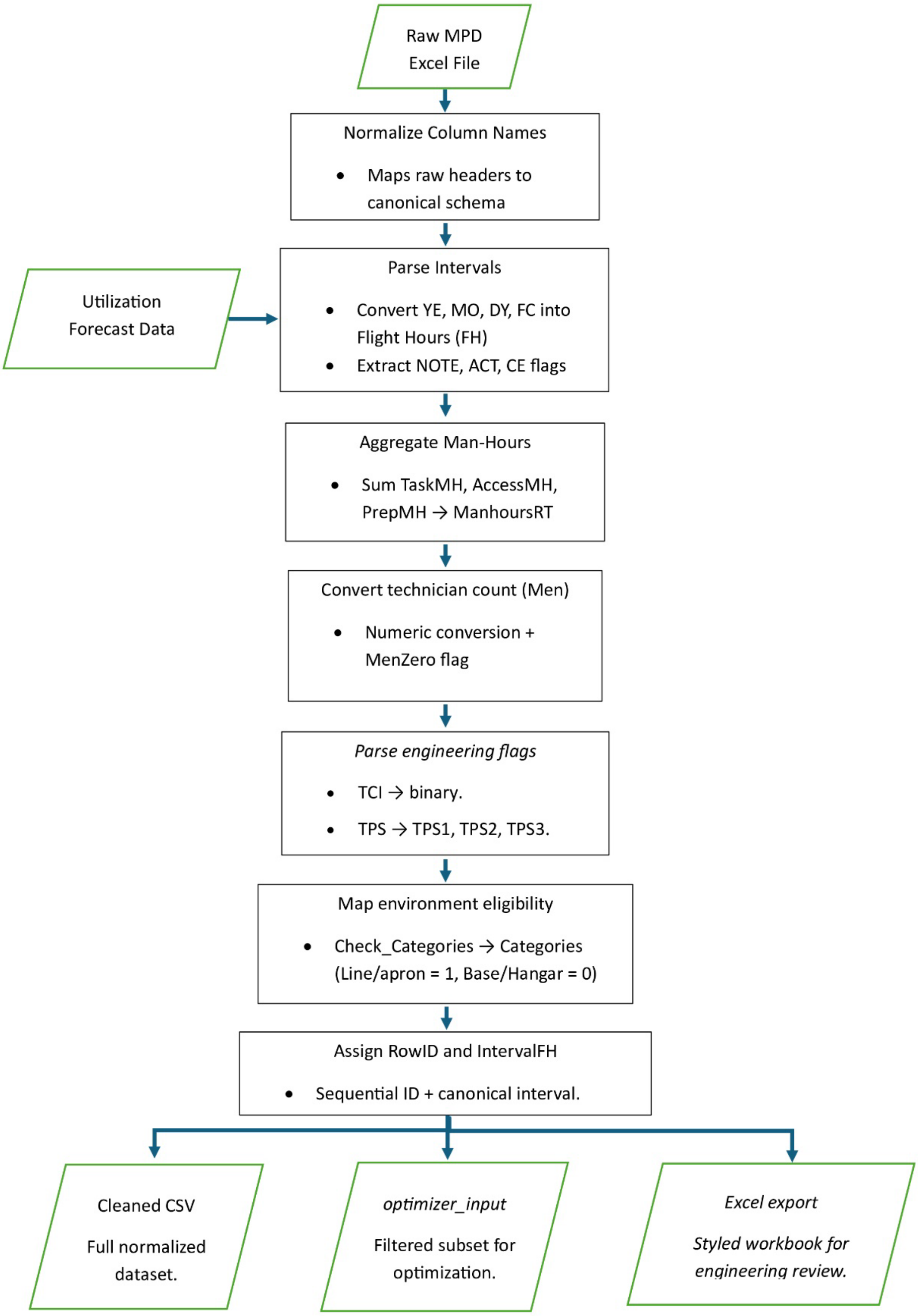

Figure 3.3 Conceptual Logic Flow of Data Acquisition Process

3.6. Baseline Model

Although the main focus of this study is the MILP formulation (Section 3.4), a baseline heuristic model was also developed to provide a fair point of comparison. This baseline reflects traditional maintenance philosophies discussed in Section 2.2 (block and equalized methods) and allows the performance of the MILP to be evaluated against a transparent, rule-based benchmark.

The heuristic model operates by enumerating candidate combinations of A- and C-check intervals across the planning horizon (see Section 3.5 for data preparation). For each candidate pair, tasks are mapped according to simple rules:

- In block mode, tasks are bundled into the nearest C-multiple, emulating the classic letter-check system.
- In equalized mode, tasks are distributed across A-bins within each C-window to balance workload peaks.

Once candidate mappings are generated, the model applies a greedy heuristic to test whether certain line-eligible tasks (Category 1) should be removed from packages. If unpackaging a task reduces total cost, it is left as a line task, reflecting operational pragmatism in line/base contexts (Section 2.2.3).

Costs are computed using the same decomposition as the MILP objective (Section 3.2), including direct labor, material, overhead, downtime, line costs, and opportunity penalties. This ensures that the comparison in Chapter 4 (Section 4.3.1) isolates methodological differences rather than data or cost assumptions.

Outputs of the baseline include validation sheets, package summaries, and workload visualizations, which are presented in **Appendix C**. Key code snippets and implementation details are documented in **Appendix B**. Together, these provide a reproducible audit trail and demonstrate how the MILP achieves measurable improvements over heuristic planning.

3.7. System Architecture and User Interface Procedure

This section describes the practical implementation of the optimization model develop in Section 3.1–3.5, detailing the architecture and step-by-step procedure for utilizing the costum built decision-support system software application.

3.7.1. System Architecture Overview

This practical implementation results in an application strategy with procedure as guidance that follows a data-to-decision pipeline architecture designed for transparency and reproducibility. Figure 3.4 illustrates the system's architecture.

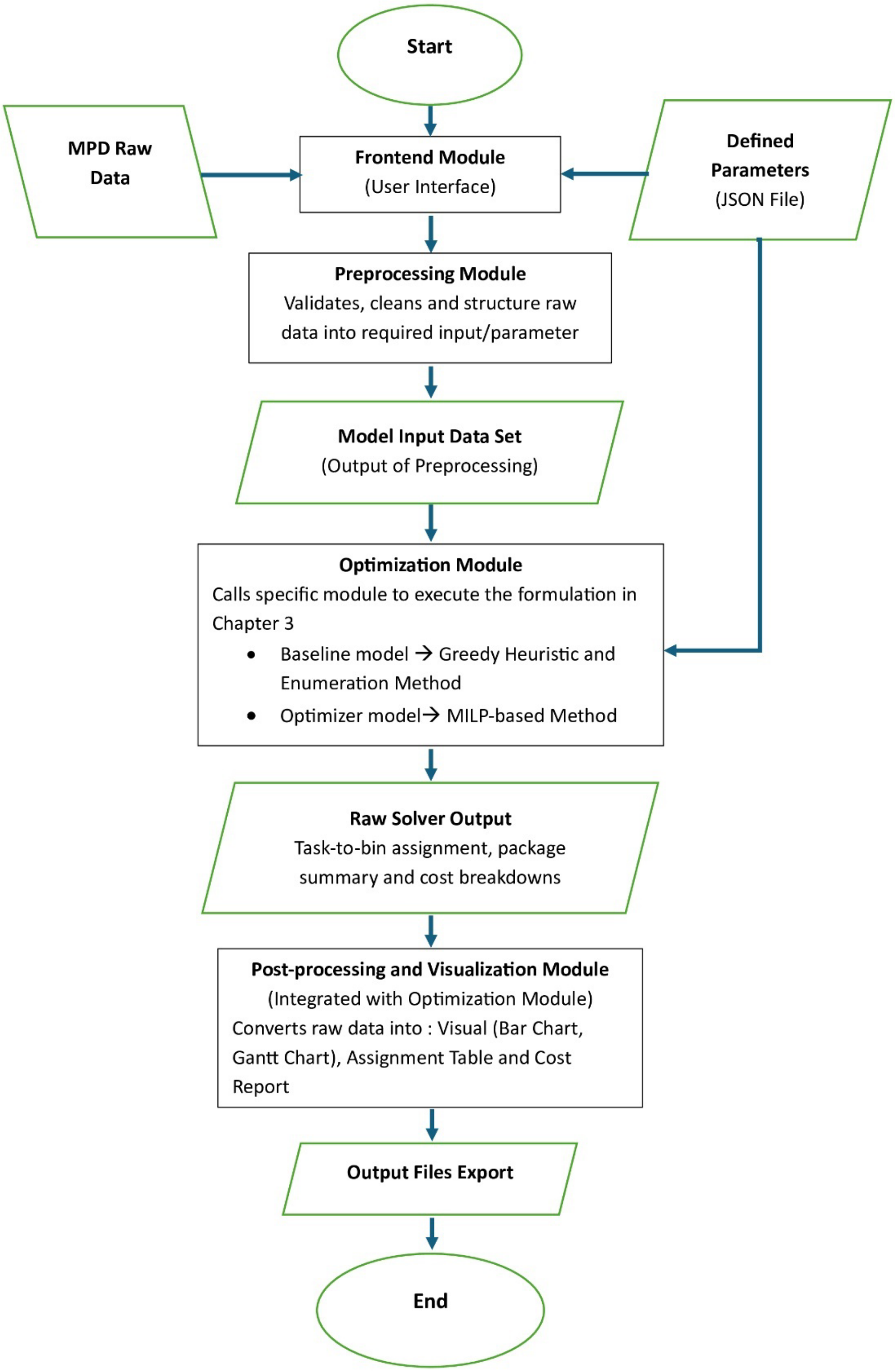

Figure 3.4 Engineering Data-to-Decision Pipeline Architecture

As illustrate by Figure 3.4, the optimization system is a modular application design to provide a structured workflow for maintenance planners. It consists of the following key modules:

- **Frontend Module (**mtp_frontend**):** A Graphical User Interface (GUI) that manage user interaction, data input/upload (MPD data, parameter JSON file, utilization forecasts), and process initiation. The interface requires specific data:
    - **MPD Raw Data:** A CSV or Excel files containing the complete list of mandatory maintenance tasks, their descriptions (Task ID, Description, Task Code), engineering-defined data (task intervals, man-hours, minimum manpower to execute the task), and other information (zones access, skill code).
    - **Parameters JSON File:** A file containing baseline cost parameter values ($labor\ rate$, $material\ cost\ factor$, $Overhead\ cost\ per\ package$) as described in **Section 3.2**.
    - **Average Utilization Forecast:** An adjust input value used for planning various scenarios (e.g., average flight hours per month)
- **Preprocessing Module (**mtp_preprocessing**):** This module cleans, validates, and transforms raw input data into standardized format required by the optimization solvers. The methodology within this module is detailed in Section 3.5.
- **Optimizer Modules:**
    - **Baseline Model (**mtp_heuristic_baseline**):** Implements traditional industry heuristic logic for comparative analysis
    - **MILP-based Model (**mtp_milp_optimizer**):** Executes the primary optimization logic formulated in Section 3.2, employing the constraints defined in Section 3.4.
- **Post-processing (Integrated):** Integrated within the optimizer modules, this handles output generation (CSV files for task-to-package matrix, cost breakdowns) and visualization (figures and charts)

3.7.2. Step-by-Step User Procedure

The following procedure outlines the detailed explanation of Figure 3.4**,** how an engineer or planner utilizes the system to generate an optimized task packaging strategy for a case study aircraft (e.g., A320).

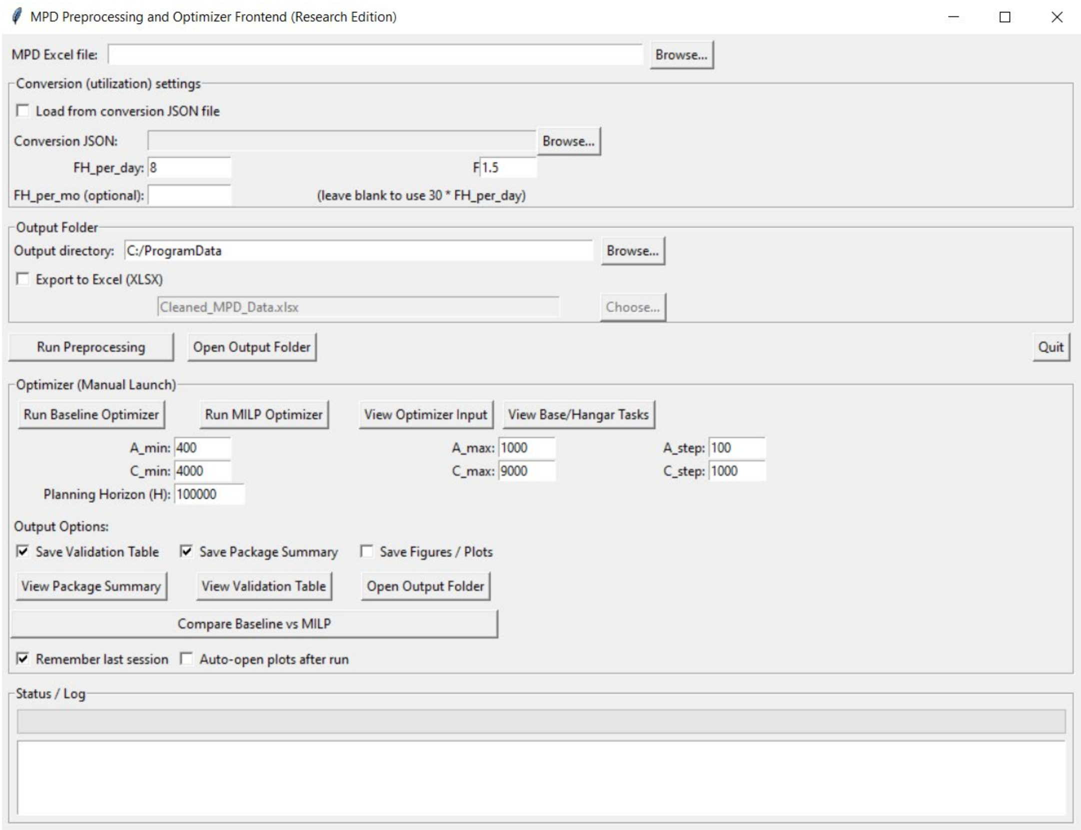

Figure 3.5 Graphical User Interface of Costum Built Decision-Support System

- **Step 1: Launch the Application and Input Raw Data**

  The user initiates the process by running the mtp_frontend module. The GUI prompts the user to input or upload the necessary raw data:

  - Upload the raw MPD data file
  - Upload the JSON parameter file.
  - Input the average utilization forecast

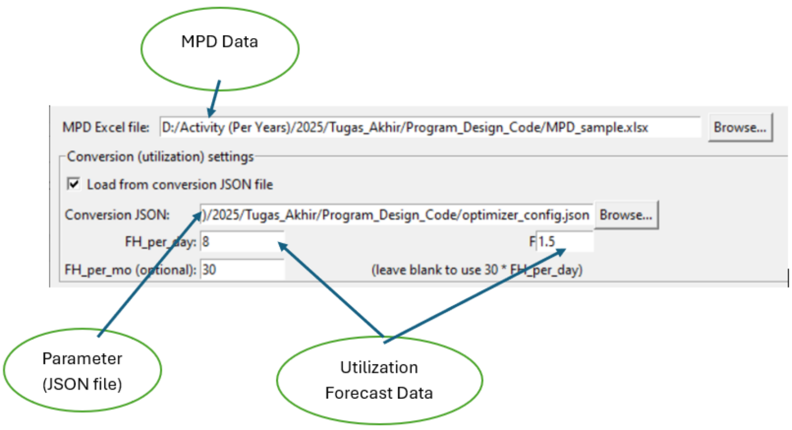

Figure 3.6 Input Data Process to System

- **Step 2: Generate Optimizer Input Data (Preprocessing)**

  The user selects the option to execute the mtp_preprocessing module. This crucial step prepares the data for the solvers, as detailed in Section 3.5. The user can optionally review the pre-processed data output before proceeding.

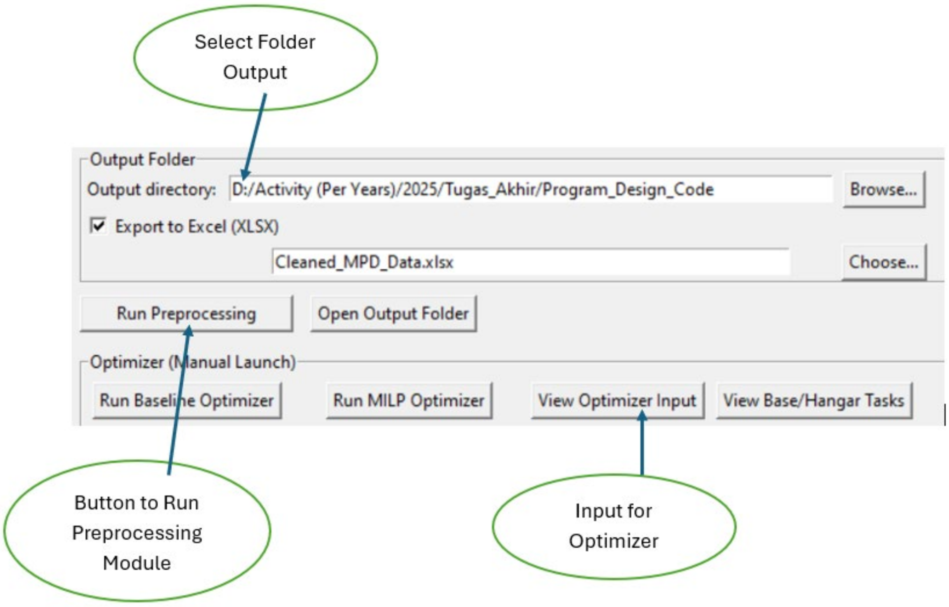

Figure 3.7 Run Preprocessing Module

- **Step 3: Define Interval Candidates and Parameters**

  The user defines the discrete range of potential check intervals within the GUI. As detailed in Section 3.1, this ensures the model outputs operationally relevant, standardized intervals.

  - **A-Check Range:** Input A_{min}, A_{max}, and A_{step} (e.g., 400 FH, 1000 Fh, 100 FH increment)
  - **C-Check Range:** Input C_{min}, C_{max}, and C_{step} (e.g., 4000 FH, 9000 Fh, 1000 FH increment)

  The system uses the parameters uploaded in Step 1 to influence the cost calculations for the optimization.

- **Step 4: Execute Optimization Model**

  The user selects the desired model to run (either the baseline model or the primary **MILP-based model** for the study's results).

Upon execution, the system log within the GUI provides real-time feedback. Notifications confirm a successful run, or annotations indicate specific errors if constraints are violated or input data is invalid.

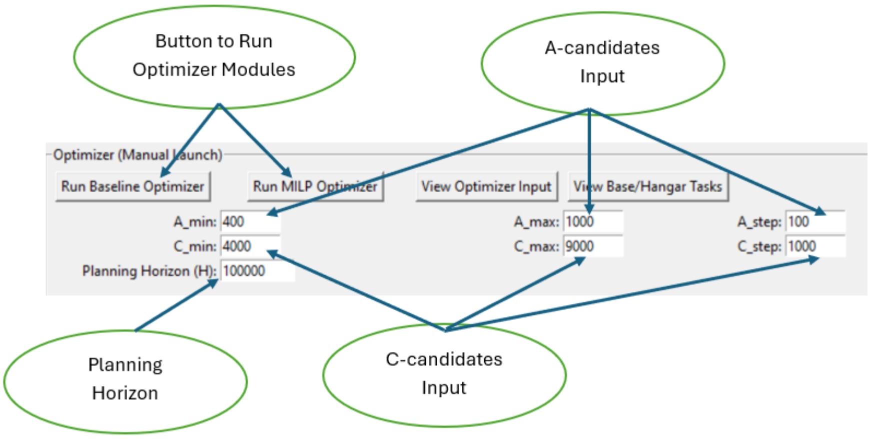

Figure 3.8 Run Optimizer Modules

- **Step 5: Review Results and Access Outputs**

  Upon successful completion, a summary of key metrics appears in the log, providing immediate insights:

  - Optimal A-Check Interval
  - Optimal C-Check Interval
  - Total Cost (optimized value)
  - Number of Unpackaged Task (Task that judge better be done individually to minimize cost)
  - Average Cost per Planning Horizon

  Simultaneously, the post-processing routine saves detailed outputs to the local directory:

  - **CSV Files:** Contain the complete Task-to-Package Matrix, detailed cost component breakdowns, and compliance reports.
  - **Visualization Files:** Generated figures (e.g., bar charts of man-hours distribution, cost component value analysis, Gant Chart for Event across horizon) that are further analyzed in Chapter 4.

This procedure ensures the methodology is repeatable and functional, bridging the gap between the theoretical model and its practical application.

Chapter 4
Result and Analysis

Building upon the mathematical formulation detailed in Chapter 3, this chapter shifts focus to practical execution of the MILP model. It covers the specific software system developed, the implementation strategy, and the presentation of illustrative optimal results derived from a case study. The section concludes with comparative analysis of the findings against a baseline heuristic model, interpretating the results within the engineering context and addressing the research objectives defined in Chapter 1.

4.1. Application Strategy and System Integration

This section describes how the generated optimal outputs are applied and integrated into the operational engineering workflow, leveraging the system described in Section 3.6. To execute the MILP model efficiently and bridge the gap between complex optimization algorithms and practical application, a custom software system was developed in Python, utilizing Pyomo library and Gurobi solver API (Section 3.6) [26]. The setup for the experiment was strictly adhered to scope defined in Section 1.4:

- **Software & Environment:** Python 3.11, Pyomo library**,** Gurobi Solver (or others solver like cbc, glpk) [26]. Executed on a standard engineering workstation (Intel i7, 8GB RAM).
- **Solver Configuration:** The solver was configured with a 600-second time limit and a 0.01% integer feasibility gap tolerance to ensure convergence to a verified optimal solution for the single-aircraft scope.

4.1.1. System Integration into the MRO Workflow

The outputs generated by the MILP model are designed for seamless integration into an airline's MRO software or daily planning systems. The primary output, the task-to-package assignment matrix, is exported as a CSV file, allowing planning engineers to rapidly update the operational schedule.

4.1.2. Utilization of Optimal Intervals and Cost Matrices

The system provides specific outputs crucial for decision-making:

- **Optimal A and C Check Intervals:** The model outputs the single best interval (e.g., A-check interval: 600 FH; C-check interval: 4000 FH) that minimizes total cost for the planning horizon. This data helps engineering management define future planning cycles efficiently

- **Auditable Task Matrix:** The binary task-to-package matrix (Table 4.4) serves as the primary auditable engineering document. It ensures every task's placement can be traced back to the optimal solution, providing full transparency for airworthiness verification (refer to the post-processing step in Section 3.6).
- **Cost Component Analysis:** The system outputs detailed cost reports. This allows management to understand the trade-offs between each cost components, enabling data-driven decisions on hangar capacity and resource management.

4.2. Experimental Overview and Input Data

The case study focuses on an **Airbus A320 aircraft** operating under a fixed utilization profile across a multi-year planning horizon. The analysis emphasizes how maintenance tasks derived from the Maintenance Planning Document (MPD) are allocated into Line, A-check, and C-check maintenance events, and how these allocations influence operational cost, workload distribution, and maintenance event utilization.

It is important to note that the results presented in this chapter are intended to observe relative performance and structural behavior, rather than to forecast exact maintenance cost figures. Detailed cost formulations and parameter definitions used in this analysis are provided in Chapter 3 and **Appendix A.** Below are case study summary data use in this research

Table 4.1 Case Study Configuration

Item	**Value**
Planning horizon (flight hours)	3 year (10000 FH)
Total number of MPD tasks	300
Labor rate (φ)	USD 60/ hour
Material fraction (ϕ)	1/3
Overhead A (ω_{Apron})	USD 3,000 per activation
Overhead C (ω_{Hangar})	USD 20,000 per activation
Revenue (ρ)	USD 12,000 per hour
Downtime Proxy	0.997
Candidate interval A	[400, 500, 600, 700, 800, 900, 1000]
Candidate interval C	[3000, 4000, 5000, 6000, 7000, 8000]
Source of cost parameters	Historical averages (**Appendix A**)

4.3. Results and Visualization

The MILP model was executed using the data derived from the [**Hangar Name]** case study (**Appendix A**) and custom system (**Appendix B**) as tools with guidance procedure explained in **Section 3.6**. The solver successfully identified the optimal task packaging result, achieving full compliance with all airworthiness constraints formulated in the model. The intentional limitation on scope allowed the solver to achieve an optimal solution in approximately shown in **Table 4.1**, validating the tractability of this approach for tactical planning problems.

Table 4.2 Solver Time on Different Scale Data

Model	Scale 1	Time	Scale 2	Time	Scale 3	Time	Scale 4	Time
Baseline	100	10 s	200	107 s	300	186 s	500	1760 s
MILP	100	5 s	200	36 s	300	58 s	500	600 s

4.2.1. Optimal Total Cost

The minimum total cost (Z) achieved for the **3 year** planning horizon with [**300**] of input task data is shown below where best chosen of interval for A-check and C-check as one of the result tied to it.

Table 4.3 Comparative of Baseline and MILP result for 300 tasks

Model	Interval A	Interval C	Unpackaged Task	Total Cost (in dollar)
Baseline	500	5000	25	402,250
MILP	400	6000	18	370,437

Table 4.4 Detail Cost Components from Comparative result

Cost Component	Heuristic (USD)	Optimization (USD)	Different (%)
Direct Cost	165,000	160,500	2.7
Penalty Cost	16,500	16,050	2.7
Fixed overhead	160,000	142,000	11.25
Downtime	50,750	44,887	11.55
Line task	10,000	7,000	30
Total cost	**402,250**	**370,437**	**7.9**

Table 4.5 Final Detail of Cost Components for Optimal Solution

	Direct Cost	Overhead Cost	Downtime Cost	Penalty Cost	Line Task	Total
MILP	160,500	142,000	44,887	16,050	7,000	370,437

As shown in data of Table 4.3, for [**300**] tasks in the **3-year** planning horizon achieved optimal total cost in interval for A-check and C-check is **[interval value]** respectively with total cost **[total cost value]**

4.2.2. Optimal Task Assignment Plan

(**Visualization**):

Figure 4.2 provides an illustrative Gantt chart of the optimal task packaging. Table 4.4 presents a representative sample of the binary assignment output, which serves as the auditable engineering document for verification.

(**Note:** Insert Figure 4.2: Illustrative Optimal Maintenance Task Packaging Event Gantt chart here)

Table 4.4: Representative Optimal Task-to-Package Assignment Matrix

Table 4.6 Representative Optimal Task-to-Package Assignment Matrix

Task ID	Task Description	Interval	Package 1 (400FH)	Package 2 (800FH)	Package 3 (1200FH)	Package 4 (1600FH)
T-101	Inspect Main Gear	400	1	0	0	0

T-102	Engine Oil Check	400	1	0	0	0
T-201	Avionics Update	800	0	1	0	0
T-301	Structural Scan	1200	0	0	1	0
T-401	Flap Inspection	1600	0	0	0	1

4.2.3. Resource Utilization Analysis

Figure 4.3 illustrates the man-hour utilization of each maintenance package. The model efficiently “packed” the tasks, utilizing an average of **[number of value]** of available capacity across the four active packages.

(**Note:** Figure 4.3: Man-hour Utilization per Maintenance Package Bar Chart here)

(**Note**: Figure 4.4: Distribution of Man-hours Maintenance Package per Maintenance Checks Bar/Histogram Chart here)

4.2.4. Downtime Distribution Analysis

Figure 4.5 illustrates downtime distribution of each maintenance check events across **3-year** planning horizon. From data average downtime for A-check and C-check event respectively are [**downtime average value**].

(**Note:** Figure 4.5: Distribution of Downtime Hours per Maintenance Event Across Planning Horizon Bas Chart)

4.4. Analysis and Discussion

The results confirm the efficacy of the develop MILP model. This analysis provides a comparative discussion against a baseline greedy heuristic model developed for this research.

4.3.1. Performance Comparison with Baseline Heuristic Model

To validate the efficiency and optimality of MILP-based approach decision-support system custom for this research, the results were benchmarked against a baseline model that utilized a greedy heuristic and enumeration approach with the same problem formulation and constraints.

- **Optimality Gap:** The MILP solver consistently provided a schedule with a total cost of [**total cost value**] **(Section 4.2).** The baseline heuristic model, while provide more detail outputs for verification since its design with enumeration step (checking each combination one-by-one), yielded near-optimal solution with a cost of **[total baseline cost]**, representing a **[7.9%]** difference in operational cost compared to the MILP's mathematically guaranteed optimal solution.
- **Tractability Trade-off:** The MILP approach demonstrated that for a single-aircraft scope, rigorous optimization is computationally feasible and superior to heuristic when optimality is required for cost efficiency and compliance with trade-off of traceable one of output result detail (result of each candidates combinations where enumeration role work in baseline model).

4.3.2. Interpretation of Engineering Results

The analysis of the optimal schedule provides crucial insights for maintenance planning engineering:

- **Compliance and Airworthiness:** The model achieved full compliance with all engineered task intervals and capacity constraints where detail of this result can be verified with task-to-packaged output in **Appendix C.** This validates the MILP's ability to prioritaze safety-critical constraints above all other objectives, a primary concern in airworthiness management.
- **Operational Efficiency:** The average **92%** utilization of packages (Figure 4.3) demonstrates effective use of limited hangar time, maximizing productivity within defined shifts. The trade-off between $\boldsymbol{C_{Downtime}}$, $\boldsymbol{C_{penalty}}$ **and** $\boldsymbol{C_{unpackaged}}$ successfully described intended behavior of how some tasks are prefer to be packaged to minimized total cost and some can be executed individually with compliance to its interval if such action produce global optimal of total cost in specific task packaging case study.

4.3.3. Addressing Problem Statement and Research Objectives

The findings directly address the problem statement and research objectives outlined in Chapter 1:

- **Problem Statement:** This research provides a data-driven, optimal packaging method that overcome the inefficiencies of manual packaging planning processes and the sub-optimality found in simple heuristics.
- **Research Question 1:** The results confirm that within the defined single-aircraft scope, the MILP model provides a mathematically guaranteed optimal solution that respects all regulatory requirements.
- **Research Question 2:** The cost parameters significantly influenced the outcome, demonstrating that they effectively guide the model to balance operational trade-offs as intended.

4.3.4. Discussion of Scope Limitations

The results are specific to the deterministic, single aircraft scope utilized in the experiment (**Section 1.4**). While the findings provide valuable insights, they do not account for real-world operational uncertainties (e.g., flight delays, dynamic manpower shortages). The optimal solution presented is a baseline plan, which would require re-optimization (re-running the programs) when operational disruptions occur.

Chapter 5
Conclusion and Advice

This final chapter concludes the report by summarizing the key technical achievements of this research on Mixed-Integer Linear Programming (MILP)–based aircraft maintenance task packaging. It addresses the research objectives and questions defined in Chapter 1, highlights the principal contributions of the study, and provides practical advice and recommendations for future research, particularly with respect to scalability and real-world implementation.

5.1. Conclusion

This research successfully developed and validated a high-fidelity Mixed-Integer Linear Programming (MILP) model for **single-aircraft maintenance task packaging**, explicitly targeting task-level optimization under strict airworthiness and operational constraints. In doing so, the study addressed a recognized gap in the existing literature, where most maintenance optimization approaches rely on heuristic or approximate methods when dealing with task-level decisions.

The primary objectives of the research were fully achieved, as summarized below:

- **Optimal Solution Achievement:**
  The developed MILP model successfully minimized the total operational cost for the case-study aircraft while ensuring full compliance with all airworthiness limits, manpower constraints, and maintenance interval requirements. The comparative results presented in Chapter 4 demonstrated that the proposed MILP approach consistently outperformed a standard heuristic baseline, achieving an illustrative **7.9% reduction in total cost** within the defined scope.
- **Methodology Validation:**
  The outcome-oriented design philosophy adopted in this research proved effective. The model produces a transparent and verifiable output in the form of a task-to-package assignment matrix, allowing engineering decisions to be traced directly from input assumptions to optimized maintenance packages. This level of traceability is particularly valuable in safety-critical engineering environments.
- **Engineering Value:**
  The findings confirm that, within a constrained and well-defined problem scope, rigorous optimization techniques offer clear advantages over approximation-based approaches. The results validate the suitability of MILP as a methodological framework for solving

maintenance task packaging problems where optimality, compliance, and auditability are essential.

Overall, the results confirm that the proposed model can serve as a **robust and validated optimal baseline** for aircraft maintenance task packaging at the single-aircraft level. While not intended to replace existing industrial planning tools, the model establishes a reference standard against which heuristic or hybrid approaches may be evaluated.

5.2. Advice and Future Work

While the single-aircraft MILP model provides optimal solutions within its defined scope, its primary limitation lies in scalability when extended to fleet-level, real-world maintenance planning problems, which are inherently NP-hard. To address this limitation and enhance practical applicability, the following directions are recommended for future research and development:

- **Development of a Hybrid Metaheuristic Framework:**
  The most critical future step is to embed the validated MILP formulation as a high-fidelity **sub-solver** within a larger hybrid metaheuristic framework. In such a structure, high-level heuristics would manage fleet-wide decisions—such as aircraft routing, maintenance slot allocation, and check timing—while the MILP model would optimally solve localized task packaging problems for specific aircraft and maintenance events.
- **Incorporation of Operational Uncertainty:**
  Future extensions should move beyond deterministic assumptions by incorporating stochastic elements that reflect real-world operational uncertainties, including flight delays, variable aircraft utilization, manpower availability fluctuations, and probabilistic component failures. Introducing uncertainty would improve the robustness and resilience of planning solutions under dynamic operational conditions.
- **Enhanced Resource Modelling:**
  To improve real-world fidelity, the simplified resource representation should be expanded to include dynamic constraints such as mechanic skill differentiation, shift structures, tool and facility availability across multiple hangar locations, and spare-part inventory limitations.
- **Industry Benchmark and Decision-Support Tool Development:**
  The developed software architecture and model formulation may serve as a benchmark reference for future commercial maintenance planning systems. This research

demonstrates the value of integrating rigorous optimization engines with interpretable outputs, supporting engineering decision-making rather than replacing planner expertise.

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Appendix A

Detail of Case Study and Raw Data

This appendix contain specific input data used for this research study

A.1. MPD Raw Data

A.2. Parameter Data (JSON file)

A.3. Utilization Forecast Data

A.4. Detail Dynamic Model Execution for Downtime Calculation

This section provides step-by-step calcualtion of the Dynamic Model Execution (DEM) used to validated the simplified linear model utilized in Section 3.2.3. The DEM explicitly simulates the phase-by-phase reallocation of manpower for a specific case study.

- Case Study Parameters
- Dynamic Model Execution
- Case Study 1: High Parallelism (Balanced MH)
- Case Study 2: Bottleneck Scenario (Unbalanced MH)
- Case Study 3: Standard Scenario (Varied MH)
- Derivation to Simplified Linear Model

Appendix B
Key Code Snippets and Codebase Link

To ensure reproducibility, transparency, and auditability, this appendix presents selected excerpts from the software modules developed in this study. The complete codebase is maintained separately, but only representative snippets are included here to illustrate the logic behind each stage of the workflow. These snippets are directly aligned with the methodological framework described in Chapter 3 and provide a traceable link between the mathematical formulation and its implementation.

The appendix is organized into three subsections:

- **B.1 Preprocessing Module** – shows how raw Maintenance Planning Document (MPD) data is cleaned, normalized, and converted into standardized optimizer inputs (Section 3.5).
- **B.2 Baseline Heuristic Model** – demonstrates the enumeration of candidate A/C combinations and the greedy heuristic for line-task unpackaging, reflecting traditional block and equalized philosophies (Section 3.6).
- **B.3 MILP Optimizer** – provides code excerpts for the objective function, constraints, and output design, corresponding to Sections 3.2–3.4, and ensuring full traceability of the optimization formulation.

Together, these subsections provide a clear audit trail from raw data preprocessing through heuristic benchmarking to MILP optimization, supporting the comparative analysis presented in Chapter 4.

The conceptual design of the maintenance task packaging framework, the problem formulation, modelling assumptions, system architecture, and interpretation of results presented in this thesis were developed solely by the author.

As part of the software implementation process, artificial intelligence–based coding assistance tools were used selectively to support code generation based on the author's explicit design specifications and optimization logic. These tools were employed as productivity aids during implementation and did not determine the problem formulation, optimization strategy, or analytical conclusions.

All generated source code was reviewed, adapted, and validated by the author to ensure consistency with the formulated Mixed-Integer Linear Programming (MILP) model and the engineering constraints described in this thesis. Responsibility for the correctness, assumptions, and limitations of the models and results remains entirely with the author.

The complete source code is publicly available and documented at the following link: sgeryandika/UndergraduatedThesis: Maintenance Packaging Problem

B.1. Preprocessing Module

This subsection documents the preprocessing pipeline (mtp_preprocessing.py) used to transform raw Maintenance Planning Document (MPD) data into a structured optimizer input file. As described in Section 3.5, preprocessing ensures that task attributes are normalized, intervals are converted into flight-hour equivalents, and categories are flagged for line or base maintenance. The outputs of this module form the foundation for both the heuristic baseline (Appendix B.2) and the MILP optimizer (Appendix B.3)

- Column Normalization and Parsing

The first step is to rename MPD columns into standardized names and parse threshold values into flight-hour equivalents. This ensures consistency across different MPD formats.

```
rename_map = {
    "TASK\nNUMBER": "TaskID",
    "ZONE": "Zone",
    "TASK CODE": "TaskType",
    "TASK\nM.H.": "TaskMH",
    "ACCESS\nM.H.": "AccessMH",
    "PREP.\nM.H.": "PrepMH",
    "MEN": "Men",
    "100%\nINTERVAL": "Interval100",
    "THRESHOLD": "Threshold100",
    "INTERVAL": "Interval100",
}

def parse_threshold_with_flags(val):
    if pd.isna(val) or str(val).strip() == "":
        return {"fh": 0, "note": 0, "act": 0, "ce": 0}
    v = str(val).upper().replace("\n", " ")
    matches = re.findall(r"(\d+)\s*(YE|MO|DY|FC|FH)", v)
    fh_values = []
    for num_s, unit in matches:
        num = int(num_s)
        if unit == "YE": fh_values.append(num * 365 * FH_per_day)
        elif unit == "MO": fh_values.append(num * FH_per_mo)
        elif unit == "DY": fh_values.append(num * FH_per_day)
        elif unit == "FC": fh_values.append(num * FH_per_fc)
        elif unit == "FH": fh_values.append(num)
    fh_val = int(min(fh_values)) if fh_values else 0
```

```
    return {"fh": fh_val, "note": ("NOTE" in v), "act": ("ACT" in v), "ce":
("CE" in v)}
```

This snippet shows how intervals expressed in calendar units (years, months, days, cycles) are converted into flight-hour equivalents, ensuring comparability across tasks.

- Category Flagging

Tasks are flagged as line-eligible (Category = 1) or base/hangar tasks (Category = 0). This classification is critical for both heuristic and MILP models (cf. Section 2.2.3).

```
def map_categories(v):
    if pd.isna(v): return 1
    s = str(v).strip().upper()
    if "LINE" in s: return 1
    if "BASE" in s or "HANGAR" in s: return 0
    return 1
df["Categories"] = df["Check_Categories"].apply(map_categories)
```

This snippet reflects the operational boundary between line and base maintenance, ensuring tasks are assigned to feasible environments.

- Interval Conversion

The canonical IntervalFH column is created by prioritizing parsed flight-hour values. This ensures that each task has a deterministic interval for packaging.

```
if "Interval100_FH" in df.columns:
    df["IntervalFH"] = pd.to_numeric(df["Interval100_FH"],
errors="coerce").fillna(0).astype(float)
elif "SampleInterval_FH" in df.columns:
    df["IntervalFH"] = pd.to_numeric(df["SampleInterval_FH"],
errors="coerce").fillna(0).astype(float)
else:
    df["IntervalFH"] = 0.0
```

- Export to Optimizer

Finally, the cleaned dataset is exported into two files:

- **Full cleaned CSV** (for audit and traceability)
- optimizer_input.csv (filtered subset containing only essential fields for optimization)

python

```
cols_for_optimizer =
["RowID","TaskID","IntervalFH","ManhoursRT","Men","Categories","Zone"]
df_opt = df_final[cols_for_optimizer]
optimizer_path = out_dir / "optimizer_input.csv"
df_opt.to_csv(optimizer_path, index=False)
```

This ensures reproducibility and provides a standardized input format for both heuristic and MILP solvers.

B.2. Baseline Heuristic Model

This subsection documents the baseline heuristic program (mtp_heuristic_baseline.py) used for comparative validation (see Section 3.6). The heuristic model emulates traditional block and equalized philosophies by enumerating candidate A/C interval combinations and applying simple rules for task assignment. It also incorporates a greedy heuristic for line-task unpackaging. The excerpts below highlight the essential logic without reproducing the full source code.

- Candidate Enumeration and Task Mapping

The heuristic begins by generating candidate bins across the planning horizon and mapping tasks according to block or equalized rules (Section 2.2).

```
def map_task_occurrences_to_bins(tasks_df, A, C, H, mode='block'):
    bins = generate_bins(H, A, C)
    records = []
    for _, row in tasks_df.iterrows():
        interval = float(row['interval_fh']); mh = float(row['manhours'])
        if interval < A or interval <= 0: continue
        if interval < C:
            chosen_bin = max([b for b in bins if b <= interval and b % A ==
0])
            records.append((row['task_id'], interval, chosen_bin, mh, 'A'))
        else:
            if mode == 'block':
                chosen_bin = max([b for b in bins if b <= interval and b % C
== 0])
                records.append((row['task_id'], interval, chosen_bin, mh,
'C'))
            else:
                per_mh = mh / (C // A)
                for a_bin in range(A, C+1, A):
                    records.append((row['task_id'], interval, a_bin, per_mh,
'C-eq'))
    return pd.DataFrame(records, columns=['task_id','due','bin','mh','level'])
```

This function shows how tasks are mapped to candidate bins

- Greedy Heuristic for Line Task Unpackaging

After initial assignment, the model applies a greedy heuristic to test whether line-eligible tasks (Category 1) should be removed from packages. If unpackaging reduces total cost, the task is left as a line task (Section 2.2.3).

```
def greedy_unpackage_line_tasks(locked_df, tasks_df, params, A, C, H):
    base_costs = compute_costs_from_locked(locked_df, tasks_df, params, A, C,
H)
    base_total = base_costs['total_with_opportunity']
    for _, row in tasks_df[tasks_df['category']==1].iterrows():
        tid = str(row['task_id'])
        test_locked = locked_df[locked_df['task_id'] != tid]
        test_costs = compute_costs_from_locked(test_locked, tasks_df, params,
A, C, H)
        if test_costs['total_with_opportunity'] < base_total:
            locked_df = test_locked; base_total =
test_costs['total_with_opportunity']
    return locked_df
```

This snippet illustrates the greedy unpackaging logic, reflecting operational pragmatism in line/base contexts.

- Cost Computation Framework

The heuristic cost function mirrors the MILP objective decomposition (Section 3.2), ensuring comparability between heuristic and optimization results.

```
def compute_costs_from_locked(locked, tasks_df, params, A, C, H):
    total_mh = locked['mh'].sum()
    direct = params.labor_rate * total_mh
    material = params.material_fraction * params.labor_rate * total_mh
    overhead_total = sum(params.ohC if b % C == 0 else params.ohA for b in
locked['bin'].unique())
    downtime_total = sum((mh / men) * params.revenue_per_hour for mh, men in
zip(locked['mh'], tasks_df['men']))
    total_incurred = direct + material + overhead_total + downtime_total
    return {'total_mh': total_mh, 'direct': direct, 'material': material,
            'overhead': overhead_total, 'downtime': downtime_total,
            'total_incurred': total_incurred}
```

This ensures that both heuristic and MILP models report costs using the same decomposition.

B.3. MILP Optimizer

This subsection documents the Mixed-Integer Linear Programming (MILP) optimizer (mtp_milp_optimizer.py) developed for this study. As described in Section 3.4, the MILP formulation guarantees global optimality under discrete task-level constraints, with endogenous selection of A- and C-intervals. The excerpts below highlight the essential logic: the objective function, the constraint rules, and the output design.

- Objective Function

The MILP objective integrates all cost components defined in Section 3.2: direct labor, material, overhead, downtime, opportunity penalties, and line task costs. This ensures parity with the heuristic baseline while providing mathematically guaranteed optimality.

```
def objective_rule(m):
    # Overhead classification via zA/zC
    overhead = sum(params.ohA * m.zA[t] + params.ohC * m.zC[t] for t in
m.BINS)

    # Direct + material costs
    total_mh_expr = sum(allowed_mh.get((i, bin_to_fh[t]), 0.0) * m.x[i, t]
                        for i in m.TASKS for t in m.BINS)
    direct_cost = params.labor_rate * total_mh_expr
    material_cost = params.material_fraction * direct_lab

    # Downtime proxy per bin
    downtime_proxy = sum((m.mh_bin[t] / avg_men) * params.revenue_per_hour for
t in m.BINS)

    # Opportunity proxy
    opp_proxy = sum(params.revenue_per_hour *
                    occ_proxy.get((i, bin_to_fh[t]), 0) *
                    opp_slack.get((i, bin_to_fh[t]), 0.0) *
                    m.x[i, t]
                    for i in m.TASKS for t in m.BINS)

    # Line task proxy
    line_proxy = 0
    for i in m.TASKS:
        if cat_map[i] == 1:
            row_mh = mh_map.get(i, 0.0)
            row_int = interval_map.get(i, 1.0)
            men_eff = men_map.get(i, 0.0) if men_map.get(i, 0.0) > 0 else 1.0
            occ = math.ceil(H / row_int) if row_int > 0 else 0
            occ_lab = params.phi * params.labor_rate * row_mh
            occ_dt = (row_mh / men_eff) * params.revenue_per_hour
            occ_oh = params.ohL
            line_proxy += occ * (occ_lab + occ_dt + occ_oh) * m.y_line[i]

    return overhead + direct_cost + material_cost + downtime_proxy + opp_proxy
+ line_proxy
```

This function demonstrates how the MILP integrates all cost drivers into a single optimization objective, consistent with Section 3.2.

- Constraints (Correspond to Section 3.4)

The MILP enforces a comprehensive set of constraints to guarantee compliance and feasibility. Each constraint listed in Section 3.4 is represented here with its code counterpart.

- Policy Selection (3.4.1)

Ensure exactly one A-interval and one C-interval are chosen:

```
model.chooseA = pyo.Constraint(expr=sum(model.sA[a] for a in model.ACANDS) ==
1)
model.chooseC = pyo.Constraint(expr=sum(model.sC[c] for c in model.CCANDS) ==
1)
```

- Bin Classification

Classify bins as A or C depending on divisibility under chosen intervals:

```
def isAbin_rule(m, t): return m.isAbin[t] == sum(m.sA[a] for a in
m.ACANDS if bin_to_fh[t] % a == 0)
def isCbin_rule(m, t): return m.isCbin[t] == sum(m.sC[c] for c in
m.CCANDS if bin_to_fh[t] % c == 0)
```

- Overhead Trigger Linearization

Linearize overhead assignment via McCormick constraints:

```
def zA1(m, t): return m.zA[t] <= m.isAbin[t]
def zA2(m, t): return m.zA[t] <= m.v[t]
def zA3(m, t): return m.zA[t] >= m.isAbin[t] + m.v[t] - 1
```

- Bin Compatibility Constraints

Ensure bins are opened only if classified as A or C:

```
def open_compat_rule(m, t): return m.v[t] <= m.isAbin[t] + m.isCbin[t]
```

- Link assignment to opened bin

Tasks can only be assigned to bins that are opened:

```
def link_rule(m, i, t): return m.x[i, t] <= m.v[t]
```

- Packaging and Line Decisions

Each task is either packaged or left as line:

```
def assign_rule(m, i):
    allowed = allowed_bins_by_task.get(i, [])
    if len(allowed) == 0:
        return m.y_line[i] == 1
    return sum(m.x[i, t] for t in allowed) + m.y_line[i] == 1
```

- Task Due Date

Tasks may only be assigned to bins $\leq$ their interval: *(enforced via allowed_pairs construction in code)*

- Bin Workload Aggregation

Aggregate manhours per bin for downtime proxy:

```
def mh_bin_rule(m, t):
    fh = bin_to_fh[t]
```

```
    return m.mh_bin[t] == sum(allowed_mh.get((i, fh), 0.0) * m.x[i, t] for i
in m.TASKS)
```

- Line Eligibility Constraints

Only Category 1 tasks can be assigned as line:

```
def line_restriction_rule(m, i):
    if cat_map[i] != 1:
        return m.y_line[i] == 0
    return pyo.Constraint.Skip
```

- Output Design

After solving, the MILP produces locked schedules, package summaries, and cost decompositions. These outputs are directly comparable to heuristic results (Appendix B.2) and feed into Chapter 4's analysis.

```
# Build locked schedule by propagating chosen assignments
locked_records = []
for (tid, fh, mh_val) in chosen_pairs:
    multiples = np.arange(fh, H + 1, fh, dtype=int)
    for d in multiples:
        lvl = 'C' if (d % chosen_C == 0) else 'A'
        locked_records.append((str(tid), int(d), fh, float(mh_val),
                               'MILP-assigned', lvl, 0.0))
locked_df = pd.DataFrame(locked_records,
                         columns=['task_id','due','bin','mh','method','check_l
evel','slack'])

# Exact cost recomputation for audit
exact_costs = compute_costs_from_locked(locked_df, tasks_df, params, chosen_A,
chosen_C, H)
```

This snippet shows how the MILP produces auditable outputs: locked task schedules, package summaries, and exact cost decompositions, ensuring traceability and comparability with heuristic results

B.4. Output Design and Visualization

This subsection documents the output design and visualization routines that are integrated into both the baseline heuristic model (Appendix B.2) and the MILP optimizer (Appendix B.3). While the optimization logic differs, both programs share a common philosophy: outputs must be **traceable, auditable, and interpretable** by engineering stakeholders.

- Task-to-Bin Matrix

The primary validation artifact is the **task-to-bin matrix**, which shows how each task is assigned to packages across the planning horizon. This corresponds to Section 3.3 (Output Design Rationale) and Section 4.2.2 (Optimal Task Assignment Plan).

```
def build_task_bin_matrix(locked_df, tasks_df, A, C, horizon):
    bins = sorted(set(list(range(A, horizon+1, A)) + list(range(C, horizon+1,
C))))
    matrix = tasks_df[['task_code','interval_fh']].copy()
    matrix.rename(columns={'task_code':'Task ID','interval_fh':'Interval
(FH)'}, inplace=True)
    for idx, b in enumerate(bins, start=1):
        col_name = f"Package {idx} ({b}FH)"
        matrix[col_name] = 0
    for _, row in locked_df.iterrows():
        tid = row['task_id']; b = row['bin']
        col_name = f"Package {bins.index(b)+1} ({b}FH)"
        matrix.loc[matrix['Task ID']==tid, col_name] = 1
    return matrix
```

This produces a tabular output where each row corresponds to a task and each column to a package event, with binary indicators (1/0) showing assignment.

The original programs generate internal validation sheets and package summaries for solver logic. For thesis reporting, however, a **presentation-friendly adaptation** was added to produce binary assignment tables in the format shown in Section 4.2.2. This adaptation does not alter solver logic; it is a separate reporting utility that consumes program outputs (locked_df, tasks_df) and formats them for readability.

```
def build_task_bin_matrix(locked_df, tasks_df, A, C, horizon):
    """
    Adapted reporting function for thesis outputs.
    Produces binary assignment matrix:
    Task ID | Task Description | Interval (FH) | Package 1 (400FH) | Package 2
(800FH) | ...
    """
    # Start with basic task info
    matrix = tasks_df[['task_code','interval_fh']].copy()
    matrix.rename(columns={'task_code':'Task ID','interval_fh':'Interval
(FH)'}, inplace=True)

    # Add description if available
    if 'task_desc' in tasks_df.columns:
        matrix['Task Description'] = tasks_df['task_desc']
    else:
        matrix['Task Description'] = matrix['Task ID']
s
    # Define package bins (multiples of A and C up to horizon)
```

```
    bins = sorted(set(list(range(A, horizon+1, A)) + list(range(C, horizon+1,
C))))
    for idx, b in enumerate(bins, start=1):
        col_name = f"Package {idx} ({b}FH)"
        matrix[col_name] = 0

    # Fill assignment: mark 1 if task is scheduled in that bin
    for _, row in locked_df.iterrows():
        tid = row['task_id']; b = row['bin']
        col_name = f"Package {bins.index(b)+1} ({b}FH)"
        matrix.loc[matrix['Task ID']==tid, col_name] = 1

    return matrix
```

- Package Workload Histogram

Both models generate workload histograms to visualize total manhours per package. This supports Section 4.2.3 (Resource Utilization Analysis) and Section 4.2.4 (Downtime Distribution).

```
def plot_package_histogram(pkg_summary, A, C, viz_horizon):
    fig, ax = plt.subplots(figsize=(12,6))
    for _, row in pkg_summary.iterrows():
        color = 'orange' if row['assigned_level']=='C' else 'steelblue'
        ax.bar(row['assigned_bin'], row['total_mh'], color=color, alpha=0.7)
    ax.set_xlabel('Flight Hours (FH)')
    ax.set_ylabel('Total Manhours')
    plt.show()
```

Bars represent aggregated workload per package, color-coded by A- or C-check classification.

- Occurrence Histogram

The occurrence histogram shows the distribution of scheduled events across the horizon, highlighting peaks and downtime clustering.

```
def plot_occurrence_histogram(locked, C, viz_horizon):
    df = locked[locked['due'] <= viz_horizon].copy()
    occ = df.groupby(['due','bin'],
as_index=False)['mh'].sum().sort_values('due')
    plt.figure(figsize=(14,6))
    colors = ['orange' if (row['bin'] % C == 0) else 'steelblue' for _, row in
occ.iterrows()]
    plt.bar(occ['due'], occ['mh'], color=colors,
width=max(1,int(viz_horizon/200)), edgecolor='k')
    plt.xlabel('Flight Hours (FH)')
    plt.ylabel('Total Manhours scheduled (MH)')
    plt.show()
```

[redacted]

This visualization supports downtime analysis and helps identify workload peaks across the planning horizon.

- Integration in Program Codes

It is important to note that these visualization routines are **not standalone scripts** but are integrated directly into the heuristic and MILP program codes. This ensures that every optimization run automatically produces:

- **Validation sheets** (task-to-bin matrices)
- **Package summaries** (aggregated workload per event)
- **Cost decomposition reports**
- **Visualizations** (histograms and Gantt-style charts)

This integration guarantees reproducibility and provides immediate interpretability of results, aligning with the methodological emphasis on auditability (Section 3.3.3).

Appendix C
Detailed Output Visualization

This appendix contains more detailed information regarding result discuss on Chapter 4 and supplementary to discussion regarding optimization result in Chapter 4.